

Non-cognate immunity proteins provide broader defenses against interbacterial effectors in microbial communities

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Abstract

Dense microbial communities, like the gut and soil microbiomes, are dynamic societies. Bacteria can navigate these environments by deploying proteins that alter foreign cells' behavior, such as interbacterial effectors. Current models suggest that adjacent sibling cells are protected by an immunity protein, as compared to toxin-antitoxin systems that act only within the effector-producing cell. A prevailing hypothesis is that immunity proteins binding to specific (cognate) protein partners is sufficient to disrupt effector function. Further, there is little-to-no crosstalk with other non-cognate effectors. In this research, we build on sporadic reports challenging these hypotheses. We show that immunity proteins from a newly defined protein family can bind and protect against non-cognate PD-(D/E)XK-containing effectors from diverse phyla. We describe the domains essential for binding and function and show that binding alone is insufficient for protective activity. Moreover, we found that these effector and immunity genes co-occur in individual human microbiomes. These results expand the growing repertoire of bacterial protection mechanisms and the models on how non-cognate interactions impact community structure within complex ecosystems.

eLife assessment

This study provides **valuable** insights into the specificity and promiscuity of toxic effector and immunity protein pairs. While the work is improved over a previous version, there are still some questions regarding the methodology used to draw certain conclusions, rendering the study somewhat **incomplete**. Nevertheless, this work will likely be of interest to microbiologists and biochemists working with toxin-antitoxin systems and effector-immunity proteins.

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Introduction

Specificity between protein-protein interactions is key for many biological processes, such as metabolism, development, and intercellular signaling. Binding to an incorrect partner or disrupted binding of the specific (cognate) protein can cause a diseased state or cell death¹. For bacterial social behaviors, cognate protein-protein interactions between cells impact organism fitness and community structure, such as excluding foreign cells². This importance of specificity between cognate partners in social behaviors remains largely unexplored. However, in other contexts, flexible (“promiscuous”) binding allows protein partners to retain their interactions when undergoing rapid mutational changes, such as during immune recognition of viral particles^{3,4}. Unknown is whether flexible binding between noncognate proteins can occur during bacterial social behaviors and thereby impact microbial communities. An expanded protective function would reveal new bacterial behaviors that influence individual fitness and community structure in dynamic ecosystems.

Microbes often exist within dense, multi-phyla communities, like the human gut microbiome, where they communicate and compete with neighbors. Bacteria can use effector-immunity protein (EI) pairs in these environments to gain advantages^{5,6}. Unlike bacterial toxin-antitoxin (TA) systems in which a single cell produces both toxic and neutralizing proteins, bacteria inject cell-modifying proteins (called interbacterial “effectors”) directly into neighboring cells via several contact-dependent transport mechanisms, including the type VI secretion system (T6SS), type IV secretion system (T4SS), and contact-dependent inhibition (CDI)^{7,8}. Clonal siblings produce the matching immunity protein that modifies the effector’s activity. For lethal effectors, both clonal and non-clonal cells are negatively impacted when binding is disrupted or absent⁵. These interactions between EI pairs can shape community composition by changing bacterial fitness.

The interaction specificity between matching EI pairs has historically defined immunity protein protection, but recent studies raise doubts. Of note, EI pairs interact within a neighboring cell which creates unique restrictions for both their protection mechanisms and their evolution⁹. Currently, the predominant model is that T6SS-associated EI pairs act like a tumbler lock-and-key, where each effector protein has a single cognate partner¹⁰. Immunity proteins bind their cognate effectors, often at the active site, to neutralize effector activity^{11,12}. However, experiments with engineered proteins reveal that small amino acid sequence changes to an immunity protein can allow it to bind effectors other than its cognate partner^{13,14}. Also, the T6SS-associated effector and immunity proteins from *Salmonella enterica* subsp. *enterica* serovar Typhimurium and *Enterobacter cloacae*, which are phylogenetically close, bind each other *in vitro* and protect against the other *in vivo*¹⁵. Another example is Tde1 and Tdi1. Homologous Tdi1 immunity proteins lacking a cognate effector (i.e., “orphans”) bound and protected against the effector from a different organism¹⁶. These studies indicate that the widely used tumbler lock-and-key model does not account for the potential breadth of immunity protein protection.

We studied an EI pair in *Proteus mirabilis* to examine this prevailing model. This opportunistic pathogen resides in human and animal guts and can cause recurrent and persistent urinary tract infections¹⁷. *P. mirabilis* encodes two T6SS-dependent EI pairs (one lethal and one non-lethal) that impact collective motility and relative fitness^{18,19}. For the lethal EI pair, previously termed Idr¹⁹, the molecular functions remained unknown. Here, we characterized this EI pair and determined the critical residues for activity, leading to the identification of two protein families. We showed that proteins in the immunity protein family bind non-cognate effectors produced by bacteria from different phyla and result in altered population structures. Structure-function assays revealed that a conserved region within the C-terminus of the immunity proteins is necessary to neutralize the *P. mirabilis* effector protein. Further, we found that the flexible EI

pairs from various phyla naturally co-occur in individual human microbiomes. These findings provide compelling evidence for cross-protection and support a critical revision of the model for EI pairs, particularly in consideration of ecological significance.

Results

RdnE is a DNA nuclease and seeds a PD-(D/E)XK subfamily

To compete against other strains, *P. mirabilis* strain BB2000 requires both the *idrD* gene and the T6SS, suggesting that the *idrD*-encoded protein functions as a T6SS-associated effector¹⁸. The *idrD* gene contains an Rhs region within its N-terminus. Many Rhs-containing effectors often contain an enzymatic domain in the C-terminus^{20,21}. As a result, we investigated the function of the final 138 amino acids at IdrD's C-terminus, now renamed “RdnE” for recognition DNA nuclease effector. We measured bacterial growth using a strain derived from BB2000 that has disruptions in its native *idrD* and downstream genes¹⁹. This *P. mirabilis* culture had 1000 fewer cells per mL when engineered to overproduce RdnE *in trans* than the negative control containing the parent empty vector (**Figure 1A**). An equivalent growth pattern occurred in *Escherichia coli* cells under the same conditions (**Figure 1 - Figure supplement 1**). Thus, RdnE was lethal *in vivo*.

RdnE's initial 86 amino acids contain a PD-(D/E)XK motif, which is suggestive of nucleotide degradation. The PD-(D/E)XK superfamily includes proteins with broad functions, including effectors that degrade DNA or RNA^{22–24}. Three residues in the catalytic site—D, D/E, and K—are required for activity²⁵. Therefore, we changed the corresponding residues in RdnE (D39, E53, and K55) to alanine, separately and together. *P. mirabilis* producing these mutant proteins showed growth equivalent to the negative control lacking RdnE (**Figure 1A**). We also saw that *E. coli* cells that were producing RdnE had morphologies that were indicative of DNA damage or stress, consistent with an SOS response^{26,27}. These cells were elongated, and the DAPI-stained DNA was distributed irregularly within the cells (**Figure 1 - Figure supplement 1**). Cells producing a D39A mutant (RdnE_{D39A}) largely did not have this appearance, although a few elongated cells remained, suggesting that the D39A mutant retained partial activity (**Figure 1 - Figure supplement 1**). Therefore, the PD-(D/E)XK motif was essential for cell death.

The importance of the PD-(D/E)XK motif for activity suggested that RdnE was a nuclease, but defining its molecular target required direct analysis. Due to its lethality in *P. mirabilis* and *E. coli* cells, we synthesized RdnE with a C-terminus FLAG epitope tag using *in vitro* translation, which resulted in nanogram quantities (**Figure 1 - Figure supplement 1**). We added phage lambda DNA (methylated or unmethylated) to progressively higher RdnE protein concentrations and then performed agarose gel electrophoresis analysis. Degradation of lambda DNA occurred in the presence of RdnE, regardless of the DNA methylation state (**Figure 1B**). The RdnE_{D39A} construct caused a slight reduction in lambda DNA, while the negative control showed no DNA loss (**Figure 1B**). RdnE also caused a reduction in plasmid DNA, indicating it has endonuclease activity (**Figure 1 - Figure supplement 1**). These results revealed that RdnE caused DNA degradation *in vitro* in a PD-(D/E)XK-dependent manner.

RdnE appeared to have two different domains, as a region directly follows the PD-(D/E)XK motif. A two-domain architecture is similar to that described for DNases^{28,29}. Yet, the PD-(D/E)XK domain could also be sufficient for DNase activity of some effectors such as PoNe-containing DNases^{22,30}. Therefore, we examined whether RdnE's PD-(D/E)XK motif was sufficient for DNA degradation or whether both domains were required for activity. We made independent deletions of each potential RdnE domain. One construct deleted the first alpha helix without disturbing the catalytic residues; the other deleted the region after the PD-(D/E)XK motif, which we termed

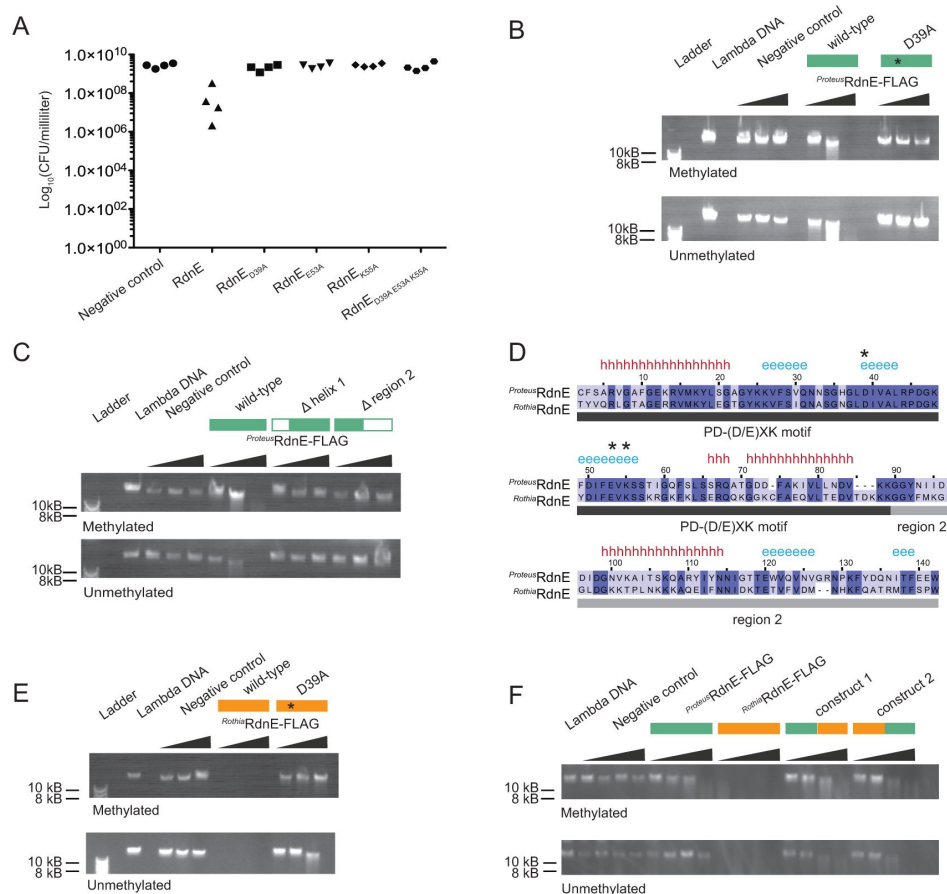


Figure 1.

RdnE homologs act as DNA endonucleases and contain interchangeable domains.

A) Cell viability (colony forming units [CFU] per mL) after protein production in swarms of *P. mirabilis* strain *idrD**, which does not produce RdnE and RdnI. Cells produced GFPmut2, RdnE, or mutant variants in the predicted PD-(D/E)XK motif: D39A, E53A, K55A, or all. **B**) *In vitro* DNA degradation assay for *Proteus* RdnE. Increasing concentrations of a negative control, *Proteus* RdnE-FLAG, or *Proteus* RdnE_{D39A}-FLAG were incubated with methylated or unmethylated lambda DNA (48,502 bp) and analyzed by gel electrophoresis. Plasmid DNA degradation is in **Figure 1 - Figure supplement 1**. **C**) *In vitro* DNA degradation assay for domain deletions of *Proteus* RdnE. The first construct removed the first alpha helix without disturbing the catalytic residues, and the second construct contained the PD-(D/E)XK motif and removed region 2. Increasing concentrations were analyzed as in **(B)**. **D**) Multiple sequence alignment between *P. mirabilis* and *R. dentocariosa* RdnE sequences. The black bar marks the PD-(D/E)XK motif, and the gray bar marks the variable region 2 domain. Conserved residues are highlighted in dark blue. Secondary structure predictions identified using Ali2D^{55,56} (h for alpha helix, e for beta sheet); the catalytic residues (stars) are noted above the alignment. **(E,F)** *In vitro* DNA degradation assay and analysis as in **(B)**. **(E)** Increasing concentrations of either a negative control, *Rothia* RdnE-FLAG, or *Rothia* RdnE_{D39A}-FLAG. **(F)** The PD-(D/E)XK motifs were swapped between the *Rothia* RdnE (orange bar) and the *Proteus* RdnE (green bar) sequences and compared to the wild-type RdnE proteins.

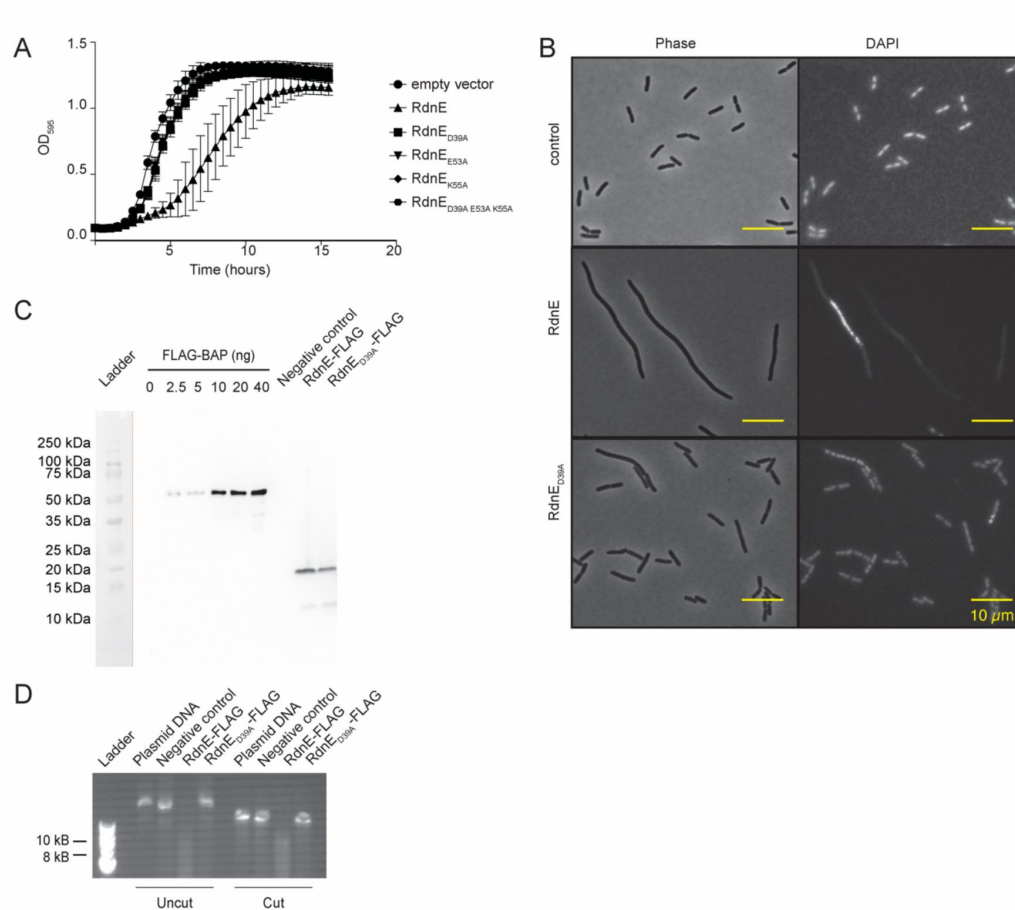


Figure 1 - Figure supplement 1.

RdnE, an endonuclease, is lethal in *Escherichia coli* and cuts plasmid DNA.

A) Growth curve of *E. coli* cells overexpressing *Proteus* RdnE or variants with mutations in the PD-(D/E)K active site. Cells were grown at 37°C for 16 hours. Optical density at 595 nm (OD₅₉₅) was measured every half hour. A control strain expressing an empty vector was used as the negative control. **B)** Micrographs of *E. coli* cells producing *Proteus* RdnE-FLAG or *Proteus* RdnE_{D39A}-FLAG, isolated during mid-logarithmic growth and imaged. DAPI was used to detect DNA within cells. Top, the empty vector as a negative control. Middle, *E. coli* producing *Proteus* RdnE from an inducible plasmid. Bottom, *E. coli* producing *Proteus* RdnE_{D39A} from an inducible plasmid. Left, phase; right, fluorescence. **C)** Anti-FLAG Western Blot for *Proteus* RdnE-FLAG and *Proteus* RdnE_{D39A}-FLAG generated by *in vitro* translation. Protein levels were determined by comparison to a standard dilution of FLAG-BAP. A negative control (DHFR) without a FLAG tag was also produced with the *in vitro* translation reaction. A vertical orange line separates the membrane where the ladder was marked with a pencil after transfer (to the left) and the membrane after western blot detection (to the right). **D)** *In vitro* DNase assay reactions on cut and uncut plasmid DNA. *In vitro* translation products of either a negative control (DHFR), *Proteus* RdnE-FLAG, or *Proteus* RdnE_{D39A}-FLAG were incubated with cut or uncut plasmid DNA and analyzed with gel electrophoresis.

“region 2.” The resulting proteins, produced via *in vitro* translation, were assayed for DNase activity as described above. The truncated proteins resulted in no loss of lambda DNA (**Figure 1C**), indicating that both domains were necessary for degradation activity.

We next asked whether RdnE homologs also act as DNA nucleases. A bioinformatics search revealed the closest RdnE homolog outside of *Proteus* was found in the Actinobacteria, *Rothia dentocariosa* C6B. *Rothia* species are inhabitants of the normal oral flora, dwelling in biofilms within the human oral cavity and pharynx³¹. The two RdnE proteins (^{*Proteus*}RdnE and ^{*Rothia*}RdnE) share approximately 55% amino acid sequence identity, mostly within the PD-(D/E)XK domain; they have similar predicted secondary structures (**Figure 1D**). Given this, we hypothesized that ^{*Rothia*}RdnE also acted as a DNA nuclease.

We analyzed ^{*Rothia*}RdnE for PD-(D/E)XK-dependent DNA nuclease activity by producing it and a predicted null mutant, ^{*Rothia*}RdnE_{D39A}, using *in vitro* translation. Samples containing the ^{*Rothia*}RdnE_{D39A} protein or a negative control had similar DNA levels (**Figure 1E**). By contrast, samples with the wild-type ^{*Rothia*}RdnE protein showed a loss of lambda DNA regardless of methylation state, indicating that ^{*Rothia*}RdnE also had DNA nuclease activity (**Figure 1E**). Given that region 2 was necessary for activity in ^{*Proteus*}RdnE but has greater amino acid sequence diversity than the PD-(D/E)XK domain between the two proteins, we queried whether domains from foreign organisms could complement one another. We exchanged region 2 between the ^{*Proteus*}RdnE and ^{*Rothia*}RdnE sequences and assayed for nuclease activity. The hybrid proteins degraded lambda DNA, unlike the negative control (**Figure 1F**), demonstrating the cross-phyla protein domains could complement one another. Altogether, these findings demonstrated that these RdnE proteins form a PD-(D/E)XK-containing DNA nuclease subfamily. This conclusion is also consistent with recent literature showing that RdnE-containing proteins (formerly IdrD-CT³²) form their own sub-clade within other PD-(D/E)XK-containing nucleases³⁰.

RdnI binds and neutralizes RdnE

As effectors have cognate immunity proteins that are often located adjacently on the chromosome, we hypothesized that *rdnI* (formerly “*idrE*”), which is the gene directly downstream of *rdnE* in *P. mirabilis* (**Figure 2A**), encodes the cognate immunity protein. RdnI did not have defined domains, and its function was unknown. We assessed RdnI’s activity using microscopic and cell growth analysis. Swarming *P. mirabilis* cells are normally elongated with DAPI-stained DNA found along the cell body (**Figure 2B**). By contrast, swarming cells producing RdnE *in trans* did not elongate, had a reduced DAPI signal, and had an accumulation of misshapen cells (**Figure 2B**). Cell shape and DNA-associated fluorescence levels returned to normal when cells concurrently produced the RdnE and RdnI proteins (**Figure 2B**). RdnI production also rescued cell growth in *E. coli* cells producing RdnE (**Figure 2 - Figure supplement 1**). These data suggested that RdnI inhibits RdnE’s lethality.

We next evaluated whether RdnI provided protection against injected RdnE within mixed communities similar to native ecosystems. We used well-established swarm competition assays, which combine one-to-one mixtures of *P. mirabilis* strains to measure dominance in two-dimensional population structures¹⁹. The control strain was wild-type strain BB2000 (herein called “BB2000”), which naturally produces RdnE and RdnI. The other was strain ATCC29906, which does not naturally produce RdnE and RdnI. These two strains formed a visible boundary between swarming monoculture colonies (**Figure 2C**). The mixed-strain colony merged with BB2000 in one-to-one competitions, demonstrating BB2000’s dominance in the two-dimensional population structure (**Figure 2C**). A similar outcome was seen when ATCC29906 produced a vector-encoded Green Fluorescent Protein (GFPmut2) under the *fla* promoter, which results in constitutive gene expression in swarming *P. mirabilis* cells^{33,34}. However, BB2000 did not outcompete ATCC29906 engineered to produce vector-encoded RdnI with a C-terminal Strep-tag II

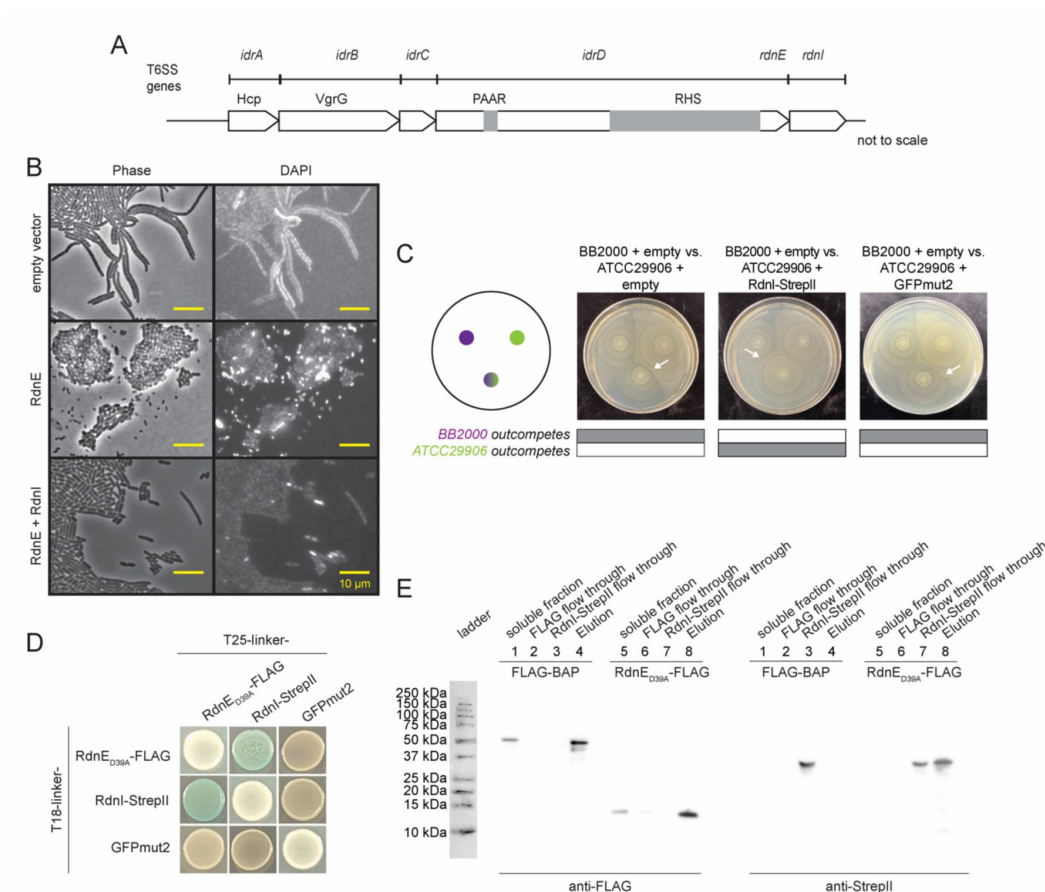


Figure 2.

RdnI binds to and protects against RdnE *in vivo* and *in vitro*.

A) Domain architecture for the *idr* locus in *P. mirabilis* strain BB2000¹⁸. At the top are genes with Pfam domains listed below them. Gray boxes denote PAAR and Rhs domains in the N-terminal region of the full-length IdrD protein. **B)** Micrographs of *P. mirabilis* strain *idrD*^{*} cells carrying an empty vector, a vector for producing RdnE, or a vector for producing RdnE and RdnI. DNA was visualized by DAPI stain. Phase, left; fluorescence, right. **C)** Swarm competition assay¹⁸ of wild-type *P. mirabilis* strain BB2000 (donor) competed against the vulnerable target, which is *P. mirabilis* strain ATCC29906 carrying an empty vector, a vector for producing RdnI-StrepII, or a vector for producing GFP, both under the *fla* promoter. Left: schematic of swarm competition assay where top left colony is BB2000, top right colony is ATCC29906 with its vector cargo, and bottom colony is a 1:1 mixture of BB2000 and ATCC29906 with its vector cargo. Gray boxes underneath indicate whether BB2000 (top) or ATCC29906 (bottom) dominate in the 1:1 mixture and white arrows point to a boundary line that forms between different strains. **D)** Bacterial two-hybrid (BACTH) assay with RdnE_{D39A}-FLAG, RdnI-StrepII, and GFPmut2. The colorimetric change was discerned in the presence of the substrate X-gal and inducer IPTG. **E)** An anti-FLAG batch co-immunoprecipitation of RdnE_{D39A}-FLAG and RdnI-StrepII. RdnE_{D39A}-FLAG or exogenous FLAG-BAP (soluble fraction) was incubated with anti-FLAG resin (FLAG flow through). RdnI-StrepII was then added to the resin (RdnI-StrepII flow through). Any proteins bound to resin were eluted with FLAG-peptide (Elution) and analyzed by anti-FLAG and anti-StrepII western blots.

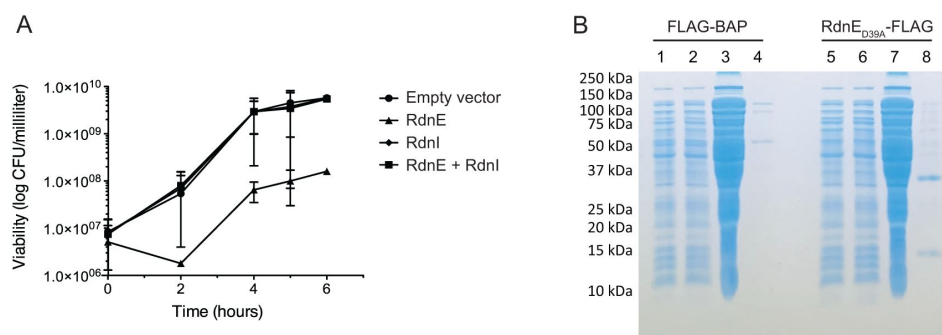


Figure 2 - Figure supplement 1.

RdnI offers protection against and binds to RdnE.

A) Viability assays of *E. coli* cells after production of RdnE, RdnI, or co-production of RdnE and RdnI within a cell. Cells were assayed for colony forming units per milliliter over a six-hour time course. **B)** The Coomassie blue-stained gel for the anti-FLAG batch co-immunoprecipitation assay results shown in the main text, **Figure 2E**.

epitope tag (“RdnI-StrepII”) under the *fla* promoter; this is visible in the merging of the mixed-strain colony with ATCC29906 (**Figure 2C**). Thus, RdnI protected cells against injected RdnE within mixed communities.

Based on the prevailing EI model, we predicted that a cognate EI pair should bind to one another, which we evaluated *in vivo* and *in vitro*. We used the attenuated mutant (RdnE_{D39A}-FLAG) for these assays because producing the wild-type RdnE protein kills cells. For *in vivo* analysis, we used bacterial two-hybrid assays (BACTH) in which the reconstitution of the T18 and T25 fragments of adenylate cyclase results in the colorimetric change to blue in the presence of the substrate, X-gal^{35,36}. Constructed vectors contained genes for RdnE_{D39A}-FLAG, RdnI-StrepII, or GFPmut2 on the C-termini of the T18 or the T25 fragment. When the reporter strain produced RdnE_{D39A}-FLAG or RdnI-StrepII with GFPmut2, the resultant yellow color was equivalent to when X-gal was absent (**Figure 2D**). There was also minimal color change when an individual protein was produced on both fragments (**Figure 2D**). However, the reporter strains made blue colonies when X-gal was present, and the cells concurrently produced RdnE_{D39A}-FLAG and RdnI-StrepII. These results indicated that RdnE and RdnI bind to each other *in vivo*.

We used batch *in vitro* co-immunoprecipitation assays to confirm the *in vivo* binding result. Separate *E. coli* strains produced either RdnE_{D39A}-FLAG or had a negative control, exogenous FLAG-BAP (*E. coli* bacterial alkaline phosphatase with a FLAG epitope tag) added to cell lysate. An anti-FLAG western blot showed both FLAG-BAP (~50 kDa) and RdnE_{D39A}-FLAG (~17 kDa) in the soluble and elution fractions. RdnI-StrepII eluted with RdnE_{D39A}-FLAG but not the negative control (**Figure 2E**). The western blot results corresponded with the Coomassie blue-stained gels (**Figure 2 - Figure supplement 1**). Overall, our data showed that *Proteus* RdnE and RdnI form a cognate EI pair with impacts on population structure. Questions about their prevalence among bacteria and their ecological relevance remained.

Expansion of the RdnE and RdnI protein families revealed similar gene architecture and secondary structures.

Gene neighborhood analysis can guide homology inference and protein comparisons. We conducted consecutive searches with BLAST³⁷ and HMMER³⁸ to identify sequences that encoded proteins with high similarity to RdnE and RdnI (**Figure 3 - Figure supplement 1**). The final list contained 21 EI pairs from a variety of phyla that are located adjacently in their respective genomes (**Table 1**). Although the genes surrounding these putative EI pairs differed, many shared mobile-associated elements, such as Rhs sequences or other similar peptide-repeat sequences (**Figure 3A**). Several gene neighborhoods had secretion-associated genes, such as the T6SS-associated *vgrG/tssI* gene and the CDI-associated *cdiB* gene. A few also included putative immunity proteins from other reported families, like immunity protein 44 (Pfam15571) in *Taylorella asinigenitalis* MCE3 and immunity protein 51 (Pfam15595) in *Chryseobacterium populi* CF314. Notably, these organisms varied widely in origin and residence. Some were from the soil rhizosphere (*Pseudomonas ogarae* and *C. populi*) and others from the human microbiome (*P. mirabilis*, *R. dentocariosa*, and *Prevotella jejuni*) (**Figure 3A**). The prevalence of these genes across the phylogenetic tree (**Figure 3B**) and the presence of secretion-associated loci in the gene neighborhoods suggested a role in cell-cell interactions and potentially community structure.

Given the diversity in species, we next examined the relationship between the various RdnE- and RdnI-like proteins and whether there was syntony between these proteins given that they are encoded adjacently on each identified genome. We constructed maximum likelihood trees to examine the relationship between the identified RdnE- and RdnI-like proteins. The RdnE and RdnI trees diverged from the species tree (**Figure 3B**). However, overall, the arrangement of RdnE- and RdnI-like proteins within the maximum likelihood trees was similar and showed syntony (**Figure 3 - Figure supplement 1**), though small differences were present (**Figure 3C**). For

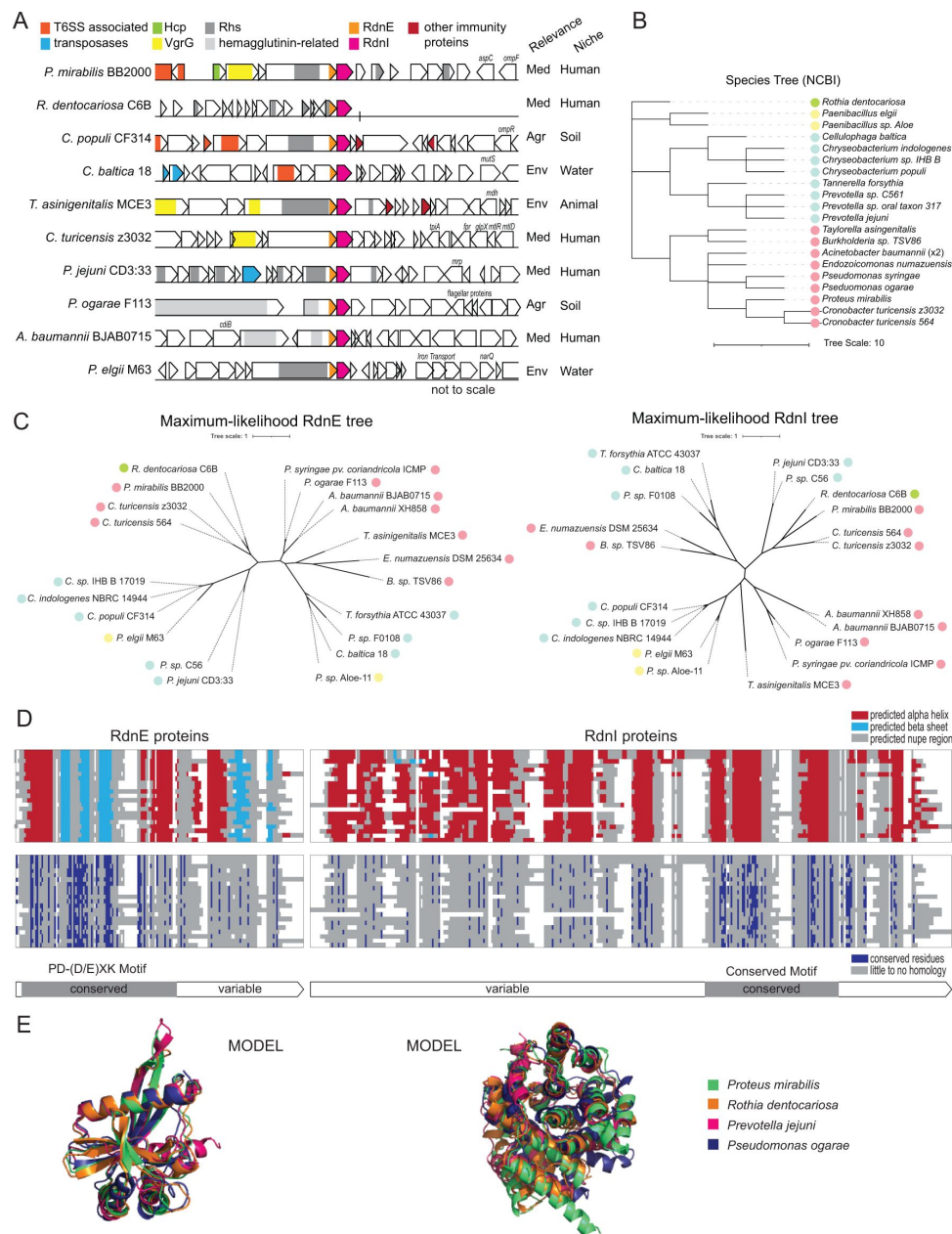


Figure 3.

RdnE and RdnI protein families share conserved residues and predicted structures.

A) Gene neighborhoods for RdnE and RdnI homologs. Listed are gene neighborhoods, relevance, and niche, which we identified using IMG/M from the Joint Genomics Institute. Colors highlight conserved function/genes (not to scale). (Agr: Agriculture, Med: Medical, Env: Environmental), and the site of isolation. **B)** Phylogenetic tree based on NCBI taxonomy. Scale is located below the graph. The colored circles represent phyla (green: Actinobacteriota; yellow: Firmicutes; blue: Bacteroidota; pink: Proteobacteria). **C)** Unrooted maximum likelihood trees of the RdnE (left) and RdnI (right) homologs. Trees were created with RaxML⁷⁰, and the scale is annotated below. The colored circles represent phyla (same as in **B**). **D)** Protein alignments overlaid with either predicted secondary structures (top) or conserved residues (bottom) of the RdnE and RdnI homologs. MUSCLE alignments⁶³ are highlighted by secondary structures (red: alpha helices, light blue: beta sheets), or conserved residues (dark blue). White represents gaps in the protein alignment. The bars below mark the predicted conserved and variable domains. **E)** Alignments of AlphaFold2 predictions for RdnE and RdnI sequences from *P. mirabilis* (green), *R. dentocariosa* (orange), *P. jejuni* (magenta), and *P. ogarae* (dark blue). Structures were generated using ColabFold⁴⁰ and aligned using PyMol.

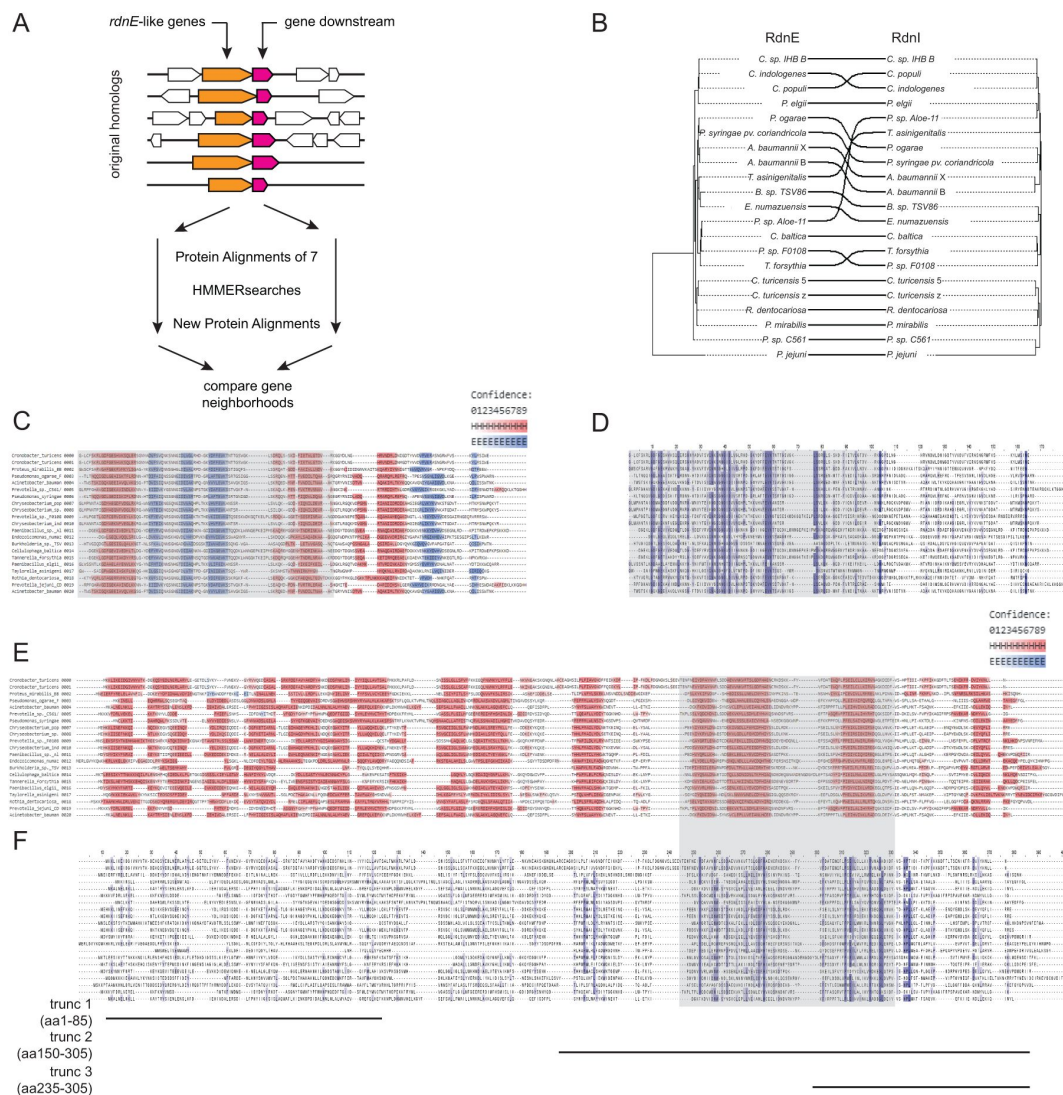


Figure 3 - Figure supplement 1.

RdnE and RdnI protein families show conserved structures.

A) Diagram detailing the methodology used for identifying sequences homologous to RdnE and RdnI. Seven homologs of *Proteus* RdnE (orange) found using BLAST³⁷ with their corresponding downstream genes (pink) were aligned and used as seed for a sequential search using HMMER search³⁸ and the Ensembl database⁶⁴. Gene neighborhoods were analyzed for genomes with adjacent *rdnE* and *rdnI* genes. **B**) Tanglegram⁷¹ of the RdnE and RdnI protein families from the 21 sequences. On the left is the maximum-likelihood tree for the RdnE protein family on the right is the maximum-likelihood tree for the RdnI protein family. Black lines match effector and immunity pairs from the same species. **C-F**) MUSCLE alignment⁶³ of RdnE (**C** and **D**) and RdnI (**E** and **F**) protein families highlighted with either predicted secondary structure predictions (**C** and **E**) or conserved residues (**D** and **F**). Secondary structures were predicted using Ali2D^{65,66} and are shaded by confidence. Predicted α -helices in pink; β -strands in light blue. Conserved residues were highlighted (dark blue) using Jalview⁶³. Black lines underneath mark the truncated variants of RdnI described in the main text **Figure 4**.

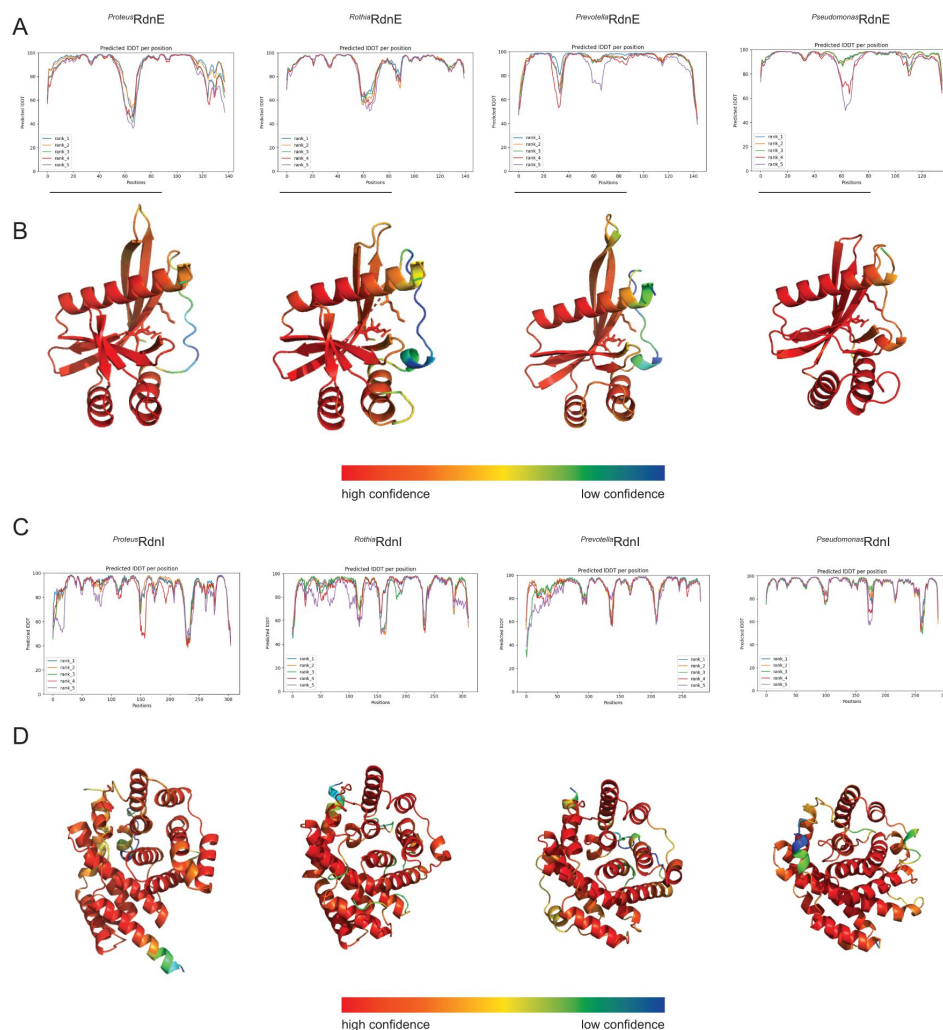


Figure 3 - Figure supplement 2.

AlphaFold2 predictions for RdnE and RdnI homologs.

A) Confidence scores (pIDDT) for AlphaFold2^{39,40} predictions for RdnE sequences from *P. mirabilis*, *R. dentocariosa*, *P. jejuni*, and *P. ogarae*. The confidence score (y-axis) for each residue (x-axis) are graphed for the five ranked models (rank 1: blue, rank 2: orange, rank 3: green, rank 4: red, rank 5: purple). **B)** RdnE AlphaFold2 rank 1 models. Models were colored by confidence scores. Red indicates high confidence (90-100%) while blue indicates low confidence (30-50%). **C)** Confidence scores for AlphaFold2 predictions of RdnI homologs from *P. mirabilis*, *R. dentocariosa*, *P. jejuni*, and *P. ogarae*. **D)** We colored the AlphaFold2 rank 1 model, including both the BB2000 sequence and natural variant, by confidence scores where red is high confidence (90-100%) and blue is low confidence (30-50%).

Genus Species Strain	Phylum	Isolation Location	RdnE JGI unique ID	RdnI JGI unique ID
<i>Acinetobacter baumannii</i> BJAB0715	Proteobacteria	Fresh water	2562302616	2562302617
<i>Acinetobacter baumannii</i> XH858	Proteobacteria	Human sputum	2686809281	2686809282
<i>Burkholderia</i> sp. TSV86	Proteobacteria	Water	2766166119	2766166118
<i>Cellulophaga baltica</i> 18	Bacteroidota	Water	2815949879	2815949878
<i>Chryseobacterium indologenes</i> NBRC 14944	Bacteroidota	Human trachea	2565567985	2565567984
<i>Chryseobacterium populi</i> CF314	Bacteroidota	Soil rhizosphere	2511231970	2511231971
<i>Chryseobacterium</i> sp. IHB B 17019	Bacteroidota	Soil undefined subtype	2686963654	2686963655
<i>Cronobacter turicensis</i> 564	Proteobacteria	Human undefined subtype	2532469359	2532469360
<i>Cronobacter turicensis</i>	Proteobacteria	Human blood	646327905	646327906

Table 1

RdnE and RdnI homolog species.

z3032		culture		
<i>Endozoicomonas numazuensis</i> DSM 25634	Proteobacteria	Marine sponge	2574519540	2574519541
<i>Paenibacillus elgii</i> M63	Firmicutes	Hot spring	2744846532	2744846531
<i>Paenibacillus</i> sp. Aloe-11	Firmicutes	Soil rhizosphere	2549870597	2549870598
<i>Prevotella jejuni</i> CD3:33	Bacteroidota	Human intestine biopsy	2804797915	2804797916
<i>Prevotella</i> sp. C561	Bacteroidota	Human respiratory tract	2514485316	2514485315
<i>Prevotella</i> sp. F0108	Bacteroidota	Human oral cavity	647936965	647936966
<i>Proteus mirabilis</i> BB2000	Proteobacteria	Human intestine biopsy	2546214711	2546214712
<i>Pseudomonas ogarae</i> F113	Proteobacteria	Soil rhizosphere	2511826458	2511826457
<i>Pseudomonas syringae</i>	Proteobacteria	Undefined	2714543877	2714543878

Table 1 (continued)

<i>pv. Coriandricola</i> ICMP 12471				
<i>Rothia dentocariosa</i> C6B	Actinobacteriota	Human oral cavity	2611822673	2611822674
<i>Tannerella forsythia</i> ATCC 43037	Bacteroidota	Human oral cavity	2512371376	2512371377
<i>Taylorella asinigenitalis</i> MCE3	Proteobacteria	Mammal reproductive system	2511725471	2511725470

Table 1 (continued)

example, *P. mirabilis* and *R. dentocariosa* proteins shared more similarities than to those from more closely related genera. These results are consistent with the potential horizontal gene transfer reported for other EI pairs⁷.

Given these results, we reasoned that the domain architectures and amino acid diversity could reveal functions for the two families. When we examined the predicted secondary structures of the RdnE-like proteins, they were conserved despite differences in the amino acid sequences (Figure 3D, Figure 3 - Figure supplement 1). The RdnE proteins showed two distinct domains, as with the *Proteus* and *Rothia* results (Figure 1): the PD-(D/E)XK region followed by a sequence variable region (region 2). Further, the AlphaFold2-generated^{39,40} predicted tertiary structures were consistent with PD-(D/E)XK folds (three β -sheets flanked by two α -helices, $\alpha/\beta/\alpha$) found in other proteins (Figure 3E, Figure 3 - Figure supplement 2)²⁵. These findings suggested that domains in RdnE are conserved across diverse phyla and further confirm that the sequences seed a distinct PD-(D/E)XK subfamily.

While immunity proteins within a family have diverse overall amino acid sequences, conserved secondary structures and some conserved residues are common in some immunity protein families. Indeed, they are often used to characterize these families⁴¹. We found that while the RdnI proteins shared minimal primary amino acid sequence identity, they were predicted to contain several α -helices (Figure 3D, Figure 3 - Figure supplement 1) and had similar AlphaFold2-predicted tertiary structures (Figure 3E, Figure 3 - Figure supplement 2). We also discovered a region with three alpha-helices and several conserved residues, which we named the “conserved motif” (Figure 3D). The RdnI conserved motif might be a key domain for seeding this novel immunity protein family.

Binding flexibility in RdnI allows for cross-species protection

We deployed a structure-function approach to determine the conserved motif’s role in *Proteus* RdnI’s activity. Analysis using AlphaFold2^{39,40} and ConSurf⁴² revealed seven highly conserved residues within this region; four of these (Y197, H221, P244, E246) clustered together within the AlphaFold2 structure and are identical between sequences (Figures 4A). In a sequence-optimized (SO) RdnI, we independently changed each of these four residues to alanine and discovered that each alanine-substituted variant behaved like the wildtype and inhibited RdnE activity (Figure 4 - Figure supplement 1). We then replaced all seven residues (Y197, S235, K258, and the original four) with alanine (*Proteus* RdnI_{7mut}-StrepII) and found that, unlike wild-type *Proteus* RdnI-StrepII, this construct was not protective in swarm competition assays (Figure 4B). However, the *Proteus* RdnI_{7mut}-StrepII mutant still bound RdnE_{D39A}-FLAG in bacterial two-hybrid assays (Figure 4C). Therefore, the seven residues in the conserved motif are critical for RdnI’s neutralizing function but dispensable for binding RdnE.

Given that the conserved motif and nearby regions are likely involved in protective activity, we queried for potential functions in the remainder of the RdnI protein. We engineered variants that were either (1) the first 85 amino acids, (2) amino acids 150 to 305, which contained an intact conserved motif, or (3) amino acids 235 to 305, which contained the last alpha helix of the conserved motif (Figure 3 - Figure supplement 1). None of these constructs protected against RdnE’s lethality *in vivo* during the swarm competition assay (Figure 4D), demonstrating that the entire protein is likely essential for function. However, the variant containing the first 85 amino acids of *Proteus* RdnI was the only construct to bind *Proteus* RdnE, indicating that the N-terminal region is sufficient for binding between this *P. mirabilis* EI pair (Figure 4E). Thus, our data suggests that binding is necessary but not sufficient for neutralization. Also, the inhibitory activity might reside within the second half of RdnI. As the prevailing model defines cognate-specificity by binding activity, our structure-function results for RdnE (Figure 1F) and RdnI (Figure 4E) suggest that this model does not fully explain the complex interactions between effectors and immunity proteins.

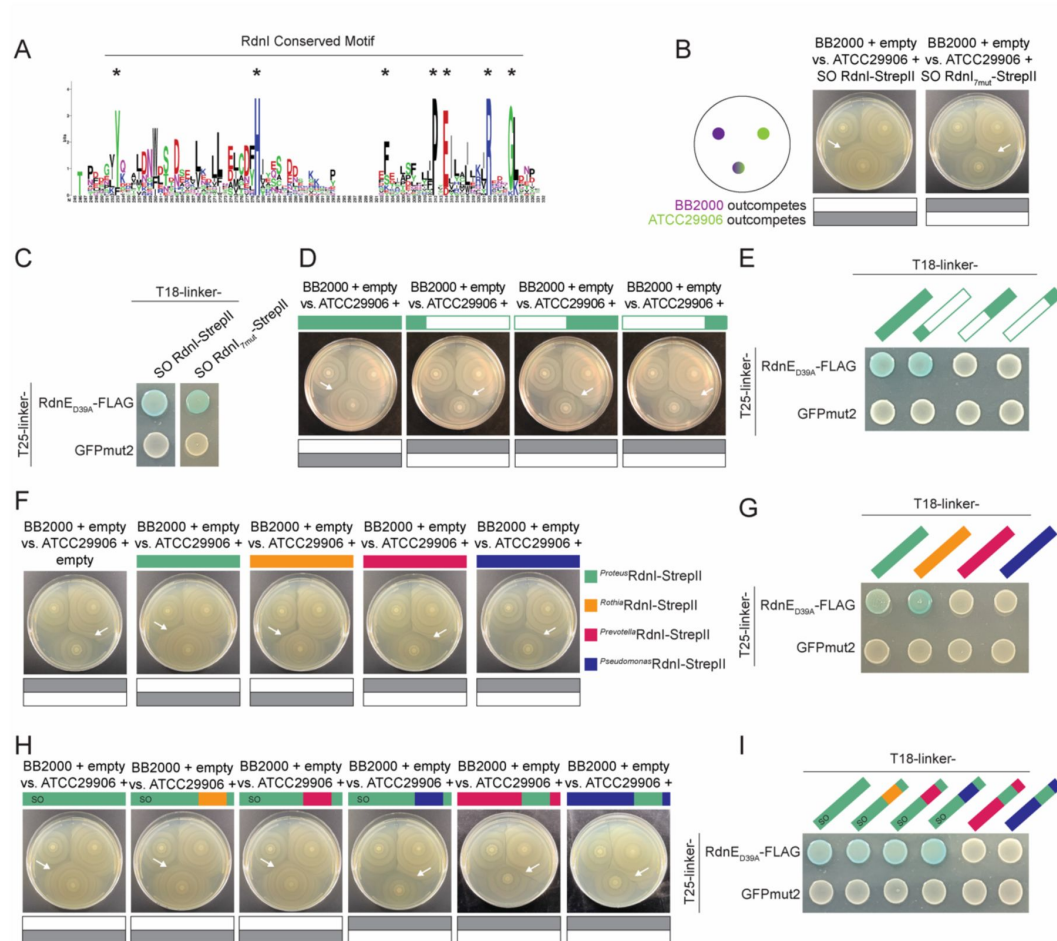


Figure 4.

The RdnI protein family can offer cross-protection due to an interchangeable conserved domain that is critical for function.

A) Sequence logo of the RdnI's conserved motif. Stars indicate the seven analyzed residues. **B)** Swarm competition assay with ATCC29906 producing either RdnI-StrepII or RdnI_{7mut}-StrepII, which contains mutations in all seven conserved residues. We used a sequence-optimized (SO) RdnI protein that had a higher GC% content and an identical amino acid sequence for ease of cloning. Left: schematic of swarm competition assay as in [Figure 2](#). Gray boxes indicate which strain dominated over the other. White arrows point to the boundary formed between different strains. **C)** BACTH assay of RdnE_{D39A}-FLAG with SO RdnI-StrepII or RdnI_{7mut}-StrepII. GFPmut2 was used as a negative control. **D)** Swarm competition assay with ATCC29906 expressing either the wild-type RdnI or a RdnI truncation. The three truncations were in the first alpha helix (amino acids 1-85), the second half of RdnI (amino acids 150-305), and the end of the protein (amino acids 235-305). **E)** BACTH assay of RdnE_{D39A}-FLAG with wild-type RdnI and the three RdnI truncations. **F)** Swarm competition assay with ATCC29906 expressing foreign RdnI proteins. **G)** BACTH assay of RdnE_{D39A}-FLAG with each of the foreign RdnI proteins. GFPmut2 was used as a negative control. **H)** Swarm competition assay with ATCC29906 producing SO RdnI with swapped conserved motifs. **I)** BACTH assay of RdnE_{D39A}-FLAG with SO RdnI with swapped conserved motifs. Colored bars denote RdnI-StrepII proteins from *P. mirabilis* (green), *R. dentocariosa* (orange), *P. jejuni* (magenta), or *P. ogarae* (dark blue).

Single mutations in the RdnI conserved motif do not alter protective function.

	BB2000 + empty vs. ATCC29906 + empty	BB2000 + empty vs. ATCC29906 + <i>SO_{J-Protein}RdnI</i>	BB2000 + empty vs. ATCC29906 + <i>SO_{J-Protein}RdnI_{Tmdl}</i>	BB2000 + empty vs. ATCC29906 + <i>SO_{J-Protein}RdnI_{Y187A}</i>	BB2000 + empty vs. ATCC29906 + <i>SO_{J-Protein}RdnI_{K221A}</i>	BB2000 + empty vs. ATCC29906 + <i>SO_{J-Protein}RdnI_{P244A}</i>	BB2000 + empty vs. ATCC29906 + <i>SO_{J-Protein}RdnI_{E245A}</i>
	 BB2000 outcompetes ATCC29906	 BB2000 outcompetes ATCC29906	 BB2000 outcompetes ATCC29906	 BB2000 outcompetes ATCC29906	 BB2000 outcompetes ATCC29906	 BB2000 outcompetes ATCC29906	 BB2000 outcompetes ATCC29906

RdnI protein levels are similar under constitutive *fla* promoter in *P. mirabilis*.

Western blot and Coomassie-stained gel analysis of protein expression in ATCC29906 and Rdn1-ST strains. The Western blot (left) shows anti-StrepII staining, and the Coomassie-stained gel (right) shows total protein loading. The lanes are labeled as follows:

- soluble fraction:** ATCC29906 + Proteinase K, ATCC29906 + Rdn1-ST + Proteinase K, ATCC29906 + Rdn1-ST + Pseudomonas, ATCC29906 + Rdn1-ST + 20 ng GFP-StrepII.
- whole cell extract:** ATCC29906 + Proteinase K, ATCC29906 + Rdn1-ST + Proteinase K, ATCC29906 + Rdn1-ST + Pseudomonas, ATCC29906 + Rdn1-ST + 20 ng GFP-StrepII.

The Western blot shows a strong band at approximately 25 kDa in the soluble fraction lanes, indicating successful protein expression and solubilization. The Coomassie-stained gel shows consistent protein loading across all lanes, with major bands at approximately 25 kDa and 37 kDa.

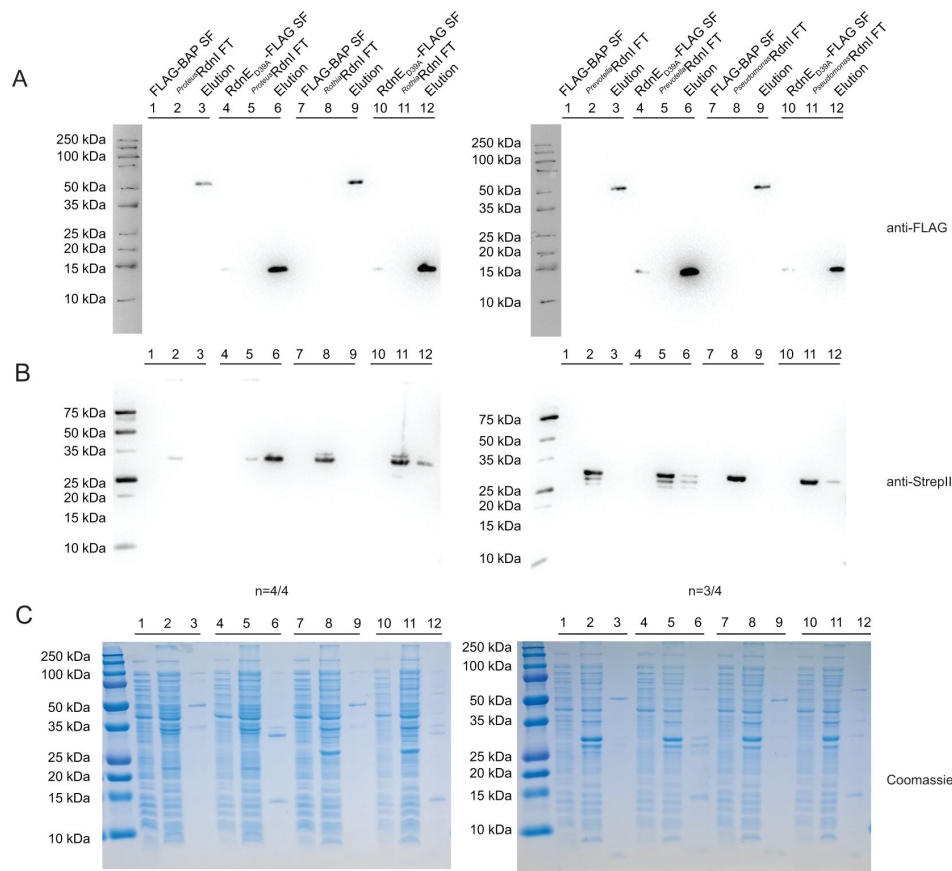


Figure 4 - Figure supplement 3.

Anti-FLAG co-IPs reveal mixed binding results for foreign immunity protein

Anti-FLAG co-immunoprecipitation assay between *Proteus* RdnE_{D39A}-FLAG and the RdnI-StrepII proteins from *P. mirabilis*, *R. dentocariosa*, *P. jejuni*, or *P. ogarae*. RdnE_{D39A}-FLAG was incubated with anti-FLAG resin (FLAG soluble fraction). RdnI-StrepII containing lysate was then added to the resin (RdnI-StrepII flow through). Proteins bound to resin were then eluted with FLAG-peptide (elution). The negative control was exogenous FLAG-BAP. Samples were incubated with either anti-FLAG antibodies (**A**), anti-StrepII antibodies (**B**), or were stained with Coomassie blue (**C**).

Therefore, we explored the relationship between non-cognate RdnE and RdnI proteins from various phyla. We first asked whether non-cognate RdnI immunity proteins could protect against injected *Proteus* RdnE. Using the swarm competition assays, we competed BB2000 against ATCC29906 engineered to produce vector-encoded RdnI homologs from *P. mirabilis*, *R. dentocariosa*, *P. jejuni*, or *P. ogarae* (*Proteus* RdnI-StrepII, *Rothia* RdnI-StrepII, *Prevotella* RdnI-StrepII, and *Pseudomonas* RdnI-StrepII, respectively) under the *fla* promoter (**Figure 2**). BB2000 dominated the swarm when ATCC29906 produced GFPmut2, *Prevotella* RdnI-StrepII, or *Pseudomonas* RdnI-StrepII (**Figure 4F**). However, ATCC29906 outcompeted BB2000 when making *Proteus* RdnI-StrepII or *Rothia* RdnI-StrepII (**Figure 4F**). Expression levels of the transgenic RdnI proteins in ATCC29906 were similar (**Figure 4 - Figure supplement 2**). Further, the RdnI immunity proteins from *Proteus* and *Rothia* consistently bound *Proteus* RdnE_{D39A} in the *in vivo* and *in vitro* assays; the *Prevotella* and *Pseudomonas* variants did not (**Figure 4G**, **Figure 4 - Figure supplement 3**). The binding to and protection of *Rothia* RdnI against *Proteus* RdnE demonstrated that cross-protection between non-cognate EI pairs from different phyla is possible, provides a fitness benefit during competition, and influences community structure.

While overall the EI pairs showed syntony with each other (**Figure 3 - Figure supplement 1**), amino acid changes can have critical impacts on whether there are specific or flexible interactions between non-cognate protein pairs⁴. Therefore, we next evaluated which region(s) of RdnI contributes to cross-protection. We first moved the conserved motif of the three foreign RdnI homologs into *Proteus* RdnI-StrepII and measured neutralizing activity using swarm competition assays and binding activity using BACTH. The conserved motifs from *Rothia* and *Prevotella* were sufficient to preserve *Proteus* RdnI's neutralizing (**Figure 4H**) and binding functions (**Figure 4I**). However, the conserved motif from *Pseudomonas* was not sufficient to neutralize *Proteus* RdnE (**Figure 4H**), even though the construct still bound *Proteus* RdnE_{D39A} (**Figure 4I**). We then moved the *Proteus* conserved motif into the RdnI variants from *Prevotella* and *Pseudomonas*. These *Prevotella* and *Pseudomonas* hybrid proteins did not protect against *Proteus* RdnE in the swarm competition assay (**Figure 4H**) or bind to it in the BACTH assay (**Figure 4I**), indicating that the conserved motif is not required for binding but, alone, is insufficient to confer protection. Thus, RdnI-like immunity proteins containing this conserved motif can protect against non-cognate effector proteins if binding has been established.

RdnE and RdnI proteins from diverse phyla are present in individual human microbiomes

Our findings revealed that immunity proteins such as RdnI could provide a broader protective umbrella for a cell beyond inhibiting the effector proteins of their siblings. If so, one would expect to find evidence of RdnE and RdnI homologs from different phyla in the same environment or microbial community. We tested this hypothesis by analyzing around 500,000 publicly available microbiomes (metagenomes) for the specific *rdnE* and *rdnI* gene sequences examined in **Figure 4** (**Figure 5A**). 2,296 human and terrestrial metagenomes contained reads matching with over 90% identity to these *rdnE* sequences (**Figure 5B**). We used this cutoff to ensure that each nucleotide sequence queried in the metagenomes closely matched experimentally characterized reference sequences. As a control, we applied a 70% identity threshold, which would retain related but more divergent sequences. We saw similar patterns with a total ~ 2% change in the number of genomes per category (**Figure 5 - Figure supplement 1**). The reads mapped to the expected niche for each organism, underscoring the presence of the genes encoding these specific effector proteins in naturally occurring human-associated microbiomes.

The *rdnE* and *rdnI* genes from various human-associated bacteria occurred concurrently in individual human oral and, to a lesser extent, gut metagenomes. The *rdnE* and *rdnI* genes from *Rothia* and *Prevotella* co-occurred in approximately 5% of the metagenomes analyzed (**Figure 5C**, **Figure 5 - Figure supplement 2**). Stringent detection parameters were utilized, so the true number could be higher. We then compared the abundance of *rdnI* to *rdnE* reads, since

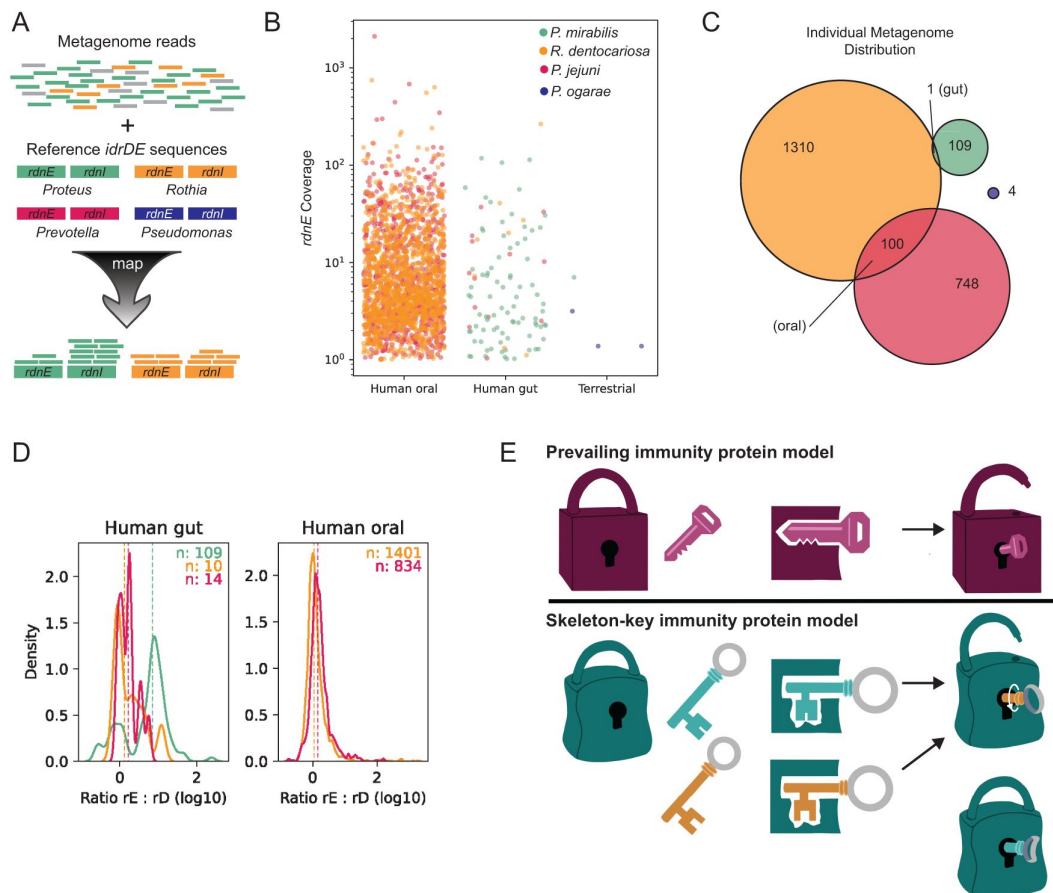


Figure 5.

The RdnI protein family has the potential for broader protection within oral and gut microbiomes.

A) Methodology used to identify *rdnE* and *rdnI* genes in publicly available metagenomic data. Metagenomes were mapped against sequences with a stringency of 90%. “Coverage” denotes the average depth of short reads mapping to a gene in a single sample. Colors represent *rdnE* and *rdnI* from *P. mirabilis* (green), *R. dentocariosa* (orange), *P. jejuni* (magenta), or *P. ogarae* (dark blue). **B)** The experimentally tested *rdnE* gene sequences from different organisms (colors) are found in thousands of human-associated metagenomes. Each dot represents a single sample’s coverage of an individual *rdnE* gene, note $\log_{10}$ -transformed y-axis. Only samples with $>1\times$ coverage are shown. **C)** Euler diagram showing the number of samples with co-occurring *rdnE* genes from different taxa (colors). **D)** Kernel density plot of the ratio of *rdnI* to *rdnE* coverage. The ratio of *rdnI* to *rdnE* was defined as $\log_{10}(I/E)$ where I and E are the mean nucleotide’s coverage for *rdnI* and *rdnE*, respectively. The distribution of ratios was summarized as a probability density function (PDF) for each taxon (color) in each environment (subpanel). Here, the y-axis (unitless) reflects the probability of observing a given ratio (x-axis) in that dataset. The colored numbers in the top right of each panel show the number of metagenomes above the detection limit for both *rdnE* and *rdnI* for each taxon. Dashed vertical lines represent the median ratio. **E)** Skeleton-key model for immunity protein protection. Top, the current prevailing model for T6SS immunity proteins is that protection is defined by necessary and sufficient binding between cognate effectors (locks) and immunity proteins (keys). Bottom, a proposed, expanded model: multiple immunity proteins (skeleton-keys) can bind a single effector due to a flexible (promiscuous) binding site. Protection is a two-step process of binding and then neutralization.

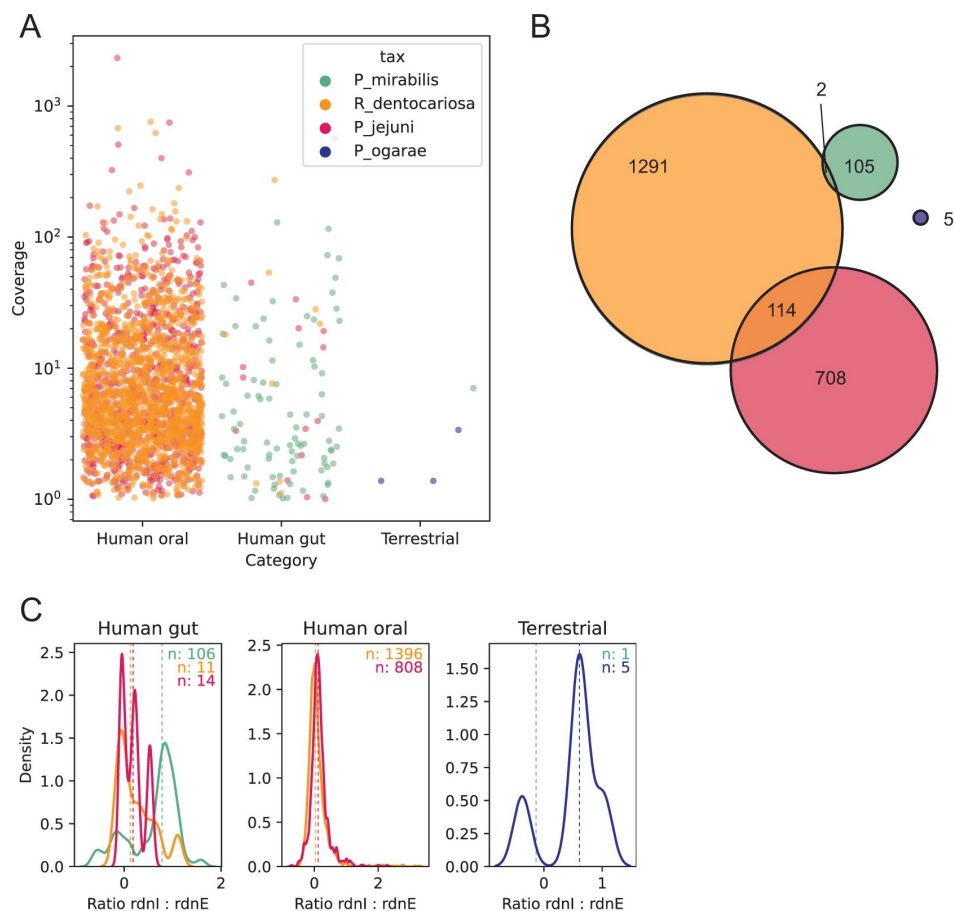


Figure 5 - Figure supplement 1.

Metagenomic analysis with a 70% stringency revealed similar patterns in *RdnE* and *RdnI* localization.

The same metagenomic analysis as described in [Figure 5](#) was used but had a lower stringency (70% instead of 90% identity to the *rdnE* and *rdnI* sequences). **A**) Each dot represents a single sample's coverage of an individual *rdnE* gene, note $\log_{10}$ -transformed y-axis. Only samples with $>1\times$ coverage are shown. **B**) Euler diagram showing the number of samples with co-occurring *rdnE* genes from different taxa (colors). **C**) Kernel density plot of the ratio of *rdnI* to *rdnE* coverage. The ratio of *rdnI* to *rdnE* was defined as $\log_{10}(I/E)$ where I and E are the mean nucleotide's coverage for *rdnI* and *rdnE*, respectively. The distribution of ratios was summarized as a probability density function (PDF) for each taxon (color) in each environment (subpanel). Here, the y-axis (unitless) reflects the probability of observing a given ratio (x-axis) in that dataset. The colored numbers in the top right of each panel show the number of metagenomes above the detection limit for both *rdnE* and *rdnI* for each taxon. Dashed vertical lines represent the median ratio.

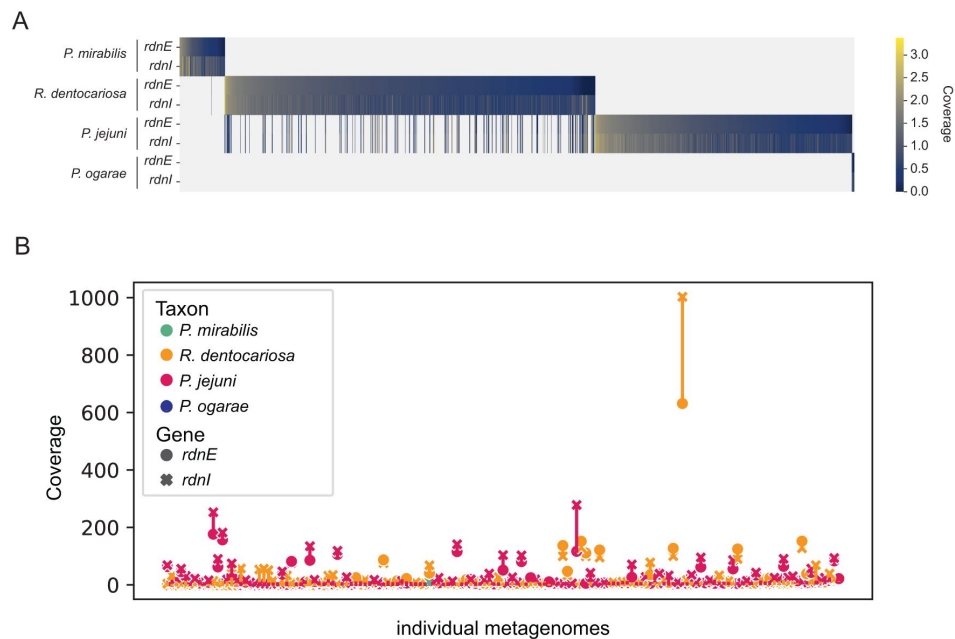


Figure 5 - Figure supplement 2.

RdnE and RdnI sequences are found in metagenomic datasets.

A) Heatmap of the log₁₀-normalized coverage of *rdnE* and *rdnI* (rows) from the focal taxa for all metagenomes (columns) where any were detected. Metagenomes are sorted by decreasing *rdnE* coverage. **B)** Span chart showing the difference in coverage between cognate *rdnE* (circles) to *rdnI* (crosses) for all taxa (colors) for metagenomes in which *rdnE-rdnI* from multiple taxa were detected.

metagenomic coverage (i.e., the number of short reads that map to a gene) approximates the underlying gene's abundance in the sampled community. In most gut samples, *rdnI* recruited more reads than *rdnE*, although there was substantial variance (**Figure 5D**). These data could indicate the presence of orphan *rdnI* genes, which is consistent with published T6SS orphan immunity alleles^{16,43,44}. These metagenomic patterns suggest that a single community can produce multiple RdnE and RdnI proteins from different phyla, thereby providing a potential for them to interact in a host environment.

Discussion

Using these results as a foundation, we propose an extension to the prevailing model of selective, cognate EI partners⁴⁵ to incorporate “EI skeleton keys.” In this revised model, flexible (“promiscuous”) binding between non-cognate effector and immunity proteins enables broader protection in mixed-species communities (**Figure 5E**). Here, we demonstrated that RdnE-RdnI binding is necessary but not sufficient to neutralize RdnE, which differs from many previously described T6SS immunity proteins. We showed that full-length RdnE, containing both its PD-(D/E)XK domain and variable C-terminal region, is a DNA-degrading endonuclease. Likewise, RdnI requires its entire protein to bind and neutralize RdnE, including a newly identified conserved C-terminal motif. Our findings point to a possible two-step mechanism for how the RdnI immunity protein works: the N-terminal variable-sequence domain mediates binding to an effector, while the C-terminal conserved domain contributes to neutralization in a not-yet-determined mechanism (**Figure 5E**). These findings have potential impacts on our understanding of immunity protein evolution, molecular functions, and microbial community structure.

The domain architectures of RdnE and RdnI suggest possible evolutionary trajectories for these EI pairs. While both require the full-length protein, RdnE and RdnI can function with residue changes within the domains and even retain activity in cross-phyla hybrid proteins. Tri1 immunity proteins also contain two domains, but these domains are associated with distinct functions. Their conserved region corresponds with the broad-acting enzymatic domain, which is independent of their cognate-specific binding domain,⁴⁶ suggesting that conserved enzymatic activity can be maintained alongside potential coevolution necessary for strict cognate pair binding. Therefore, individual domains in EI proteins may evolve independently rather than the entire protein experiencing coevolution. As such, RdnE's PD-(D/E)XK motif and RdnI's conserved motif might be maintained for activity, while the variable domains may diversify in sequence independently. Depending on the selective pressures, the variable regions could reinforce specificity between cognate EI pairs as they coevolve. Additional evolutionary analysis would reveal how the balance between specificity and flexibility evolves in EI pairs, both within domains and across the entire protein.

RdnI's potential two-step mechanism adds to a growing number of ways in which immunity proteins neutralize effector proteins. However, RdnI's neutralization mechanism remains unknown. Many crystalized structures of EI complexes show that immunity proteins can bind and occlude an effector's active site, effectively neutralizing the effector's function.^{12,47} However, some immunity proteins allosterically inhibit their effector without blocking the active site.^{48,49} Recent studies revealed more neutralization mechanisms. The Tri1 immunity protein has a conserved enzymatic function that neutralizes its effector's activity, allowing for protection against foreign effectors in addition to selective cognate-binding activity,⁴⁶ while the Tdi1 immunity protein conformationally disrupts its effector, Tde1's, binding and active sites to prevent DNA nuclease activity.¹⁶ One will need to experimentally determine the structure of the RdnE-RdnI complex is necessary to define how it neutralizes RdnI and how this molecular mechanism compares to other immunity proteins.

Regardless, our results indicate that RdnI's conserved domain is essential for protective activity and that a combination of seven highly conserved residues mediates that protection. There are several possibilities for how this domain aids neutralization, such as an ion-binding pocket, structural stability, or protein partner binding. Another possibility is that RdnI's conserved region reinforces binding to the effector, aiding non-cognate interactions or co-evolving pairs. For example, while our assays indicate the primary binding domain for RdnI is in the N-terminus, the conserved domain could reinforce an initial, transient binding interaction. Indeed, multiple binding domains have been recorded for TA systems and likely protect the cell during co-evolution⁵⁰. Binding affinities between effector and immunity proteins are not well-documented; those reported vary. For DNA nuclease colicins, non-cognate interactions have affinities in the nanomolar range whereas cognate interactions have picomolar affinities⁵¹. However, the T6SS-associated EI pair, Tde1 and Tdi1, have similar nanomolar affinities for both cognate and non-cognate orphan EI pairs¹⁶. In addition, it is unclear what equilibrium dissociation constant (K_D) for EI binding would confer protection in native systems. This K_D may be especially important in the case of highly motile bacteria, such as *P. mirabilis*, that only need to survive long enough to escape. Additionally, interbacterial effectors act within a neighboring cell, which may make determining the native ratios of effector to immunity proteins challenging. It will be interesting to see how binding affinities between other EI pairs compare, both cognate and non-cognate interactions, and how these affinities relate to protection in mixed communities.

When considering the impacts on community structure, the broadened activity of RdnI proteins against RdnE effectors from multiple phyla likely increases bacterial fitness, which is advantageous in dense environments. Our analysis measured protection during a two-dimensional, swarm-structured competition, where RdnI offered a susceptible strain protection against trans-cellular RdnE delivered natively. As such, we can conclude that RdnI production increased individual cells' fitness and modified the community structure; it enabled vulnerable bacteria to inhabit previously restricted spaces. Supporting this experimental data, both gut and oral metagenomes showed evidence of multiple *rdnE-rdnI* pairs within individual samples, particularly between *Rothia* and *Prevotella*. Interestingly, the oral microbiome had roughly equivalent abundance between the effector and immunity genes, which might reflect that bacteria occupy distinct spatiotemporal niches within oral microbiomes, e.g. *R. dentocariosa* is predominantly on tooth surfaces⁵². By contrast, *rdnI* genes had greater abundance compared to *rdnE* in the gut microbiomes, which may reflect the greater diversity in member species and community structures found in the gut⁵³. Orphan immunity genes are indeed a known phenomenon in T6SS EI literature but are usually documented through single isolate sequencing, with notable exceptions such as the Ross et al. 2019 and Bosch et al. 2023 papers. This community-level assessment affirms the presence of *rdnI* orphan genes on a population scale and points to relatively widespread immunity genes in hundreds or thousands of samples.

Given the ability of immunity genes to protect against non-cognate effectors, the presence of diverse orphan *rdnI* genes hints at the ecological complexity surrounding RdnE and RdnI. This community of immunity proteins is reminiscent of the model for shared immunity proteins within an ecosystem, called a "hyper-immunity state," which was seen among colicins in wild field mice⁵⁴. In this hyper-immunity state, a set of immunity proteins shared among a community could offer an advantage against pathogens. Invading bacteria would be unable to defend themselves from certain effectors, while the community would be protected as they share the collective immunity proteins. Flexible binding like RdnI could contribute to such a "hyper-immunity state" to help a bacterial community maintain its niche.

Indeed, bacteria have a diverse set of protective measures to ward off foreign effectors in addition to flexible immunity proteins. Recent work has identified non-specific mechanisms of protection including stress-response, physical barriers, and a stronger offense¹⁰. Orphan immunity genes also exist throughout many bacterial genomes and may be a part of this system,⁵⁵ for example, orphan immunity genes offer a fitness advantage *in vitro*^{12,16} and in mouse microbiomes⁵⁶.

Flexible EI pairs are also not limited to secretion systems but are also seen among TA pairs⁵⁷ and bacteriocins^{51,58}. Our data extends the current repertoire of protection mechanisms by adding another tool: a flexible immunity protein collection, where each immunity protein acts as a skeleton key against a wider class of effectors. This flexibility is seen among orphan immunity proteins^{12,16} and for immunity proteins with cognate effectors as in this study. Flexible binding could be a general property of T6SS immunity proteins that could be useful in dense, diverse communities, like human and soil microbiomes, where contact-dependent competition using EI pairs is critical to maintain one's population. Indeed, the physical interactions between, and evolution of, effector and immunity proteins remain a rich area for new explorations.

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Materials and Methods

Bacterial strains and media

All strains are described in **Table 2**. Strains for bacterial two-hybrid assays were transformed the day before. Overnight cultures were grown aerobically at 37°C in LB (Lennox) broth³³. *E. coli* strains were plated on LB (Lennox) agar surfaces (1.5% Bacto agar); *P. mirabilis* strains were plated on LSW agar³³ for single-colony growth or 25 mL CM55 media (blood agar base agar [Oxoid, Basingstoke, England]) for swarms. When necessary 35 µg/mL kanamycin or 100 µg/mL carbenicillin was included in the media.

Plasmid construction

Plasmids were constructed according to **Table 3**. Primers and gBlocks were ordered from Integrated DNA Technologies (IDT), Coralville, IA. *P_{idrA}*-RdnE was constructed using Polymerase Chain Reaction (PCR) to amplify the last 416 bp of the *idrD* gene from BB2000 and clone it into the SacI and AgeI sites of the pBBR1-NheI vector, resulting in plasmid pAS1054. RdnE is the final 138 amino acids of IdrD (out of its total of 1581). *P_{idrA}*-*rdnE*-*rdnI* was constructed by PCR amplifying the last 416 bp of the *idrD* gene through the end of the *rdnI* gene from BB2000, resulting in the plasmid pAS1059. The gBlock and primer sequences are archived on an OSF website (<https://osf.io/scb7z/>) and were made publicly available upon publication.

We used several standard protocols for vector construction. Seamless ligation cloning extract (Slice) was adapted from Zhang et al.⁵⁹. Restriction-digest reactions were based on manufacturer's protocols. Overlap extension (SOE) PCR Amplification was adapted from Heckman 2007⁶⁰. Plasmids were transformed into OmniMax *E. coli* and confirmed using Sanger Sequencing (UC Berkeley DNA Sequencing Facility and Genewiz, South Plainfield NJ).

Strain	Strain Name	Description	Reference or Source
<i>P. mirabilis</i> BB2000 <i>idrD</i> * + pDS0062	DS349	BB2000 <i>idrD</i> ::Tn5 (Cm ^R) producing GFPmut2 under the aTc-inducible promoter	This study
<i>P. mirabilis</i> BB2000 <i>idrD</i> * + pDS0002	DS104	BB2000 <i>idrD</i> ::Tn5 (Cm ^R) producing RdnE under the aTc-inducible promoter	This study
<i>P. mirabilis</i> BB2000 <i>idrD</i> * + pDS0058	DS344	BB2000 <i>idrD</i> ::Tn5 (Cm ^R) producing RdnE _{D39A} under the aTc-inducible promoter	This study
<i>P. mirabilis</i> BB2000 <i>idrD</i> * + pDS0059	DS345	BB2000 <i>idrD</i> ::Tn5 (Cm ^R) producing RdnE _{E53A} under the aTc-inducible promoter	This study
<i>P. mirabilis</i> BB2000 <i>idrD</i> * + pDS0060	DS346	BB2000 <i>idrD</i> ::Tn5 (Cm ^R) producing RdnE _{K55A} under the aTc-inducible promoter	This study
<i>P. mirabilis</i> BB2000 <i>idrD</i> * + pDS0061	DS347	BB2000 <i>idrD</i> ::Tn5 (Cm ^R) producing RdnE _{D39A} E53A K55A under the aTc-inducible promoter	This study

Table 2

List of strains used in this study.

<i>E. coli</i> MG1655 + pBBR1-NheI	DS068	MG1655 carrying empty vector	This study
<i>E. coli</i> MG1655 + pDS0002	DS151	MG1655 producing RdnE under the aTc-inducible promoter	This study
<i>E. coli</i> MG1655 + pDS0058	DS336	MG1655 producing RdnE _{D39A} under the aTc-inducible promoter	This study
<i>E. coli</i> MG1655 + pDS0059	DS337	MG1655 producing RdnE _{E53A} under the aTc-inducible promoter	This study
<i>E. coli</i> MG1655 + pDS0060	DS338	MG1655 producing RdnE _{K55A} under the aTc-inducible promoter	This study
<i>E. coli</i> MG1655 + pDS0061	DS339	MG1655 producing RdnE _{D39A E53A K55A} under the aTc-inducible promoter	This study
<i>P. mirabilis</i> BB2000 <i>idrD</i> * + pDS0003	DS092	BB2000 <i>idrD</i> ::Tn5 (Cm ^R) co-producing RdnE followed by RdnI under the aTc- inducible promoter	This study
<i>E. coli</i> MG1655 + pDS0003	DS170	MG1655 co-producing RdnE followed by RdnI under the aTc-inducible promoter	This study
<i>P. mirabilis</i> BB2000 + pBBR1-NheI	ANS1127	BB2000 carrying empty vector	19
<i>P. mirabilis</i>	ANS1280	ATCC29906 carrying empty vector	This study

Table 2 (continued)

ATCC29906 + pBBR1-NheI			
<i>P. mirabilis</i> ATCC29906 + pAK043	AK0132	ATCC29906 producing RdnI with a C-terminal Strep-II tag with the <i>fla</i> promoter	This study
<i>P. mirabilis</i> ATCC29906 + pLMW04-gfp	AK387	ATCC29906 producing GFPmut2 with the <i>fla</i> promoter	This study
<i>E. coli</i> MG1655 + pDS0048	DS248	MG1655 producing Rdn ^{ED39A} with a C-terminal FLAG tag under an aTc-inducible promoter	This study
<i>E. coli</i> BL21(pLysS)DE3 + pAK023	AK024	BL21(pLysS)DE3 producing RdnI with a C-terminal Strep-II tag under the T7 promoter	This study
<i>P. mirabilis</i> ATCC29906 + pAK044	AK0135	ATCC29906 producing the <i>Rothia</i> RdnI with a C-terminal Strep-II tag under the <i>fla</i> promoter	This study
<i>P. mirabilis</i> ATCC29906 + pAK045	AK0138	ATCC29906 producing the <i>Prevotella</i> RdnI with a C-terminal Strep-II tag under the <i>fla</i> promoter	This study
<i>P. mirabilis</i>	AK0141	ATCC29906 producing the	This study

Table 2 (continued)

ATCC29906 + pAK046		<i>Pseudomonas</i> RdnI with a C-terminal Strep-II tag under the <i>fla</i> promoter	
<i>E. coli</i> BL21(pLysS)DE3 + pAK058	AK261	BL21(pLysS)DE3 producing <i>Rothia</i> RdnI with a C-terminal Strep-II tag under the T7 promoter	This study
<i>E. coli</i> BL21(pLysS)DE3 + pAK059	AK262	BL21(pLysS)DE3 producing <i>Prevotella</i> RdnI with a C-terminal Strep-II tag under the T7 promoter	This study
<i>E. coli</i> BL21(pLysS)DE3 + pAK060	AK263	BL21(pLysS)DE3 producing <i>Pseudomonas</i> RdnI with a C-terminal Strep-II tag under the T7 promoter	This study
<i>P. mirabilis</i> ATCC29906 + pAK063	AK318	ATCC29906 producing the recoded <i>Proteus</i> RdnI sequence with a C-terminal Strep-II tag under the <i>fla</i> promoter	This study
<i>P. mirabilis</i> ATCC29906 + pAK065	AK320	ATCC29906 producing the <i>Rothia</i> RdnI sequence (aa195-271) inserted between aa192-266 in the recoded <i>Proteus</i> RdnI with a C-terminal Strep-II tag under the <i>fla</i> promoter	This study
<i>P. mirabilis</i> ATCC29906 +	AK321	ATCC29906 producing the <i>Prevotella</i> RdnI sequence (aa170-245) inserted	This study

Table 2 (continued)

pAK066		between aa192-266 in the recoded <i>Proteus</i> RdnI with a C-terminal Strep-II tag under the <i>fla</i> promoter	
<i>P. mirabilis</i> ATCC29906 + pAK067	AK322	ATCC29906 producing the <i>Pseudomonas</i> RdnI sequence (aa181-255) inserted between aa192-266 in the recoded <i>Proteus</i> RdnI with a C-terminal Strep-II tag under the <i>fla</i> promoter	This study
<i>P. mirabilis</i> ATCC29906 + pAK086	AK381	ATCC29906 producing the recoded <i>Proteus</i> RdnI sequence (aa192-266) inserted between aa170-245 in the <i>Prevotella</i> RdnI with a C-terminal Strep-II tag under the <i>fla</i> promoter	This study
<i>P. mirabilis</i> ATCC29906 + pAK087	AK382	ATCC29906 producing the recoded <i>Proteus</i> RdnI sequence (aa192-266) inserted between aa181-255 in the <i>Pseudomonas</i> RdnI with a C-terminal Strep-II tag under the <i>fla</i> promoter	This study
<i>P. mirabilis</i> ATCC29906 + pAK064	AK319	ATCC producing the recoded <i>Proteus</i> RdnI sequence with seven alanine mutations (Y197A, H221A,	This study

Table 2 (continued)

		S235A, P244A, E246A, R254A, K258A) and a C-terminal Strep-II tag under the <i>fla</i> promoter	
OneShot OmniMax 2 T1R Competent Cells		<i>E. coli</i> strain for cloning	Thermo Fisher Scientific, Waltham, MA
MFDpir		Mu-free <i>E. coli</i> mating strain to introduce plasmids into <i>P. mirabilis</i>	69

Table 2 (continued)

Plasmid Name	Description	Cloning Method or Source
pBBR1-NheI	empty vector with pBBR1 origin.	62
pLMW04-gfp	GFPmut2 with a constitutive <i>fla</i> promoter, (pBBR1 origin, Kan resistance).	19
pDS0002	<i>rdnE</i> with the anhydrotetracycline (aTc)-inducible promoter, P _{tet} , (pBBR1 origin, Kan resistance)	gDS0005 was recombined into amplified pAS1054 by SliCE
pDS0062	<i>gfpmut2</i> with the aTc-inducible promoter, (pBBR1 origin, Kan resistance)	restriction digest using amplified <i>gfpmut2</i> and pDS0002
pDS0048	<i>rdnE</i> _{D39A} -FLAG with the aTc-inducible promoter, (pBBR1 origin, Kan resistance)	gDS0025 was recombined into pDS0002 using restriction digest
pDS0058	<i>rdnE</i> _{D39A} with the aTc-inducible promoter, (pBBR1 origin, Kan resistance)	pDS0048 was recombined into pDS0002 using restriction digest
pDS0059	<i>rdnE</i> _{E53A} with the aTc-inducible promoter, (pBBR1 origin, Kan resistance)	gDS0026 was recombined into pDS0002 using restriction digest
pDS0060	<i>rdnE</i> _{K55A} with the aTc-inducible promoter,	gDS0027 was recombined into

Table 3

Plasmids used in this study.

	(pBBR1 origin, Kan resistance)	pDS0002 using restriction digest
pDS0061	<i>rdnE</i> _{D39A, E53A, K55A} with the aTc-inducible promoter, (pBBR1 origin, Kan resistance)	gDS0028 was recombined into pDS0002 using restriction digest
pDS0034	<i>rdnE</i> -FLAG with the aTc-inducible promoter, (pBBR1 origin, Kan resistance)	gDS0023 was recombined into amplified pDS0002 using SliCE
pDS0003	<i>rdnE-rdnI</i> with the aTc-inducible promoter, (pBBR1 origin, Kan resistance)	Amplified <i>rdnE-rdnI</i> from pAS1059 was recombined into pDS0002 using SOE PCR
pAK023	<i>rdnI</i> -StreptII with the T7 promoter, (pUC, Amp resistance)	gAK001 and pDS0003 were recombined into pET17b vector using SOE PCR
pAK043	<i>rdnI</i> -StreptII with the <i>fla</i> promoter, (pBBR1 origin, Kan resistance)	<i>rdnI</i> -StreptII amplified from pAK023 was recombined into pLMW04 using restriction digest
pAK070	T25-linker- <i>rdnE</i> _{D39A} -FLAG with the IPTG-inducible <i>lac</i> promoter, (p15A, Kan resistance)	<i>rdnE</i> _{D39A} -FLAG tag amplified from pDS0048 was recombined into pKT25 using restriction digest
pAK071	T25-linker- <i>rdnI</i> with the <i>lac</i> promoter,	<i>rdnI</i> -Strept-II amplified from

Table 3 (continued)

	(p15A, Kan resistance)	pAK043 was recombined into pKT25 using restriction digest
pAK074	T18-linker- <i>rdnE</i> _{D39A} -FLAG with the <i>lac</i> promoter, (Col E1 origin, Amp resistance)	<i>rdnE</i> _{D39A} -FLAG tag amplified from pDS0048 was recombined into pUT18C using restriction digest
pAK075	T18-linker- <i>rdnI</i> -StrepII with the <i>lac</i> promoter, (Col E1 origin, Amp resistance)	<i>rdnI</i> -Strep-II tag amplified from pAK043 was recombined into pUT18C using restriction digest
pAK076	T25- <i>gfpmut2</i> with the <i>lac</i> promoter, (p15A origin, Kan resistance)	Amplified <i>gfpmut2</i> was recombined into pKT25 using restriction digest
pAK077	T18- <i>gfpmut2</i> with the <i>lac</i> promoter, (Col E1 origin, Amp resistance)	Amplified <i>gfpmut2</i> was recombined into pUT18C using restriction digest
pAK044	<i>Rothia</i> <i>rdnI</i> -StrepII with the <i>fla</i> promoter, (pBBR1 origin, Kan resistance)	gAK003 was recombined into pAK043 using restriction digest
pAK045	<i>Prevotella</i> <i>rdnI</i> -StrepII with the <i>fla</i> promoter, (pBBR1 origin, Kan resistance)	gAK004 was recombined into pAK043 using restriction digest
pAK046	<i>Pseudomonas</i> <i>rdnI</i> -StrepII with the <i>fla</i> promoter, (pBBR1 origin, Kan resistance)	gAK005 was recombined into pAK043 using restriction digest

Table 3 (continued)

pAK079	T18- <i>Rothia</i> <i>rdnI</i> -StreptII with the <i>lac</i> promoter, (pUC, Amp resistance)	gAK003 was recombined into pUT18C using restriction digest
pAK081	T18- <i>Prevotella</i> <i>rdnI</i> -StreptII with the <i>lac</i> promoter, (pUC, Amp resistance)	gAK004 was recombined into pUT18C using restriction digest
pAK083	T18- <i>Pseudomonas</i> <i>rdnI</i> -StreptII with the <i>lac</i> promoter, (pUC, Amp resistance)	gAK005 was recombined into pUT18C using restriction digest
pAK058	<i>Rothia</i> <i>rdnI</i> -StreptII with the T7 promoter, (pUC, Amp resistance)	gAK003 was recombined into pAK023 using restriction digest
pAK059	<i>Prevotella</i> <i>rdnI</i> -StreptII with the T7 promoter, (pUC, Amp resistance)	gAK004 was recombined into pAK023 using restriction digest
pAK060	<i>Pseudomonas</i> <i>rdnI</i> -StreptII with the T7 promoter, (pUC, Amp resistance)	gAK005 was recombined into pAK023 using restriction digest
pAK063	Sequence optimized <i>rdnI</i> -StreptII with the <i>fla</i> promoter, (pBBR1 origin, Kan resistance)	gAK024 was recombined into pAK043 using restriction digest
pAK065	<i>Pm/Rd</i> <i>rdnI</i> -StreptII (<i>Proteus rdnI</i> with <i>Rothia</i> conserved motif insert) with the <i>fla</i> promoter, (pBBR1 origin, Kan resistance)	gAK026 was recombined into pAK043 using restriction digest
pAK066	<i>Pm/Pi</i> <i>rdnI</i> -StreptII (<i>Proteus rdnI</i> with <i>Prevotella</i> conserved motif insert) with the <i>fla</i> promoter, (pBBR1 origin, Kan resistance)	gAK028 was recombined into pAK043 using restriction digest

Table 3 (continued)

pAK067	^{Pm/Po} <i>rdnI</i> -StreptII (<i>Proteus rdnI</i> with <i>Pseudomonas</i> conserved motif insert) with the <i>fla</i> promoter, (pBBR1 origin, Kan resistance)	gAK027 was recombined into pAK043 using restriction digest
pAK086	^{Pj/Pm} <i>rdnI</i> -StreptII (<i>Prevotella rdnI</i> with <i>Proteus</i> conserved motif insert) with the <i>fla</i> promoter, (pBBR1 origin, Kan resistance)	gAK029 was recombined into pAK043 using restriction digest
pAK087	^{Pt/Pm} <i>rdnI</i> -StreptII (<i>Pseudomonas rdnI</i> with <i>Proteus</i> conserved motif insert) with the <i>fla</i> promoter, (pBBR1 origin, Kan resistance)	gAK030 was recombined into pAK043 using restriction digest
pAK092	T18-linker-Sequence optimized ^{Proteus} <i>rdnI</i> with a C-terminal StreptII with the <i>lac</i> promoter (Col E1 origin, Amp resistance)	gAK024 was recombined into pAK075 using restriction digest
pAK093	T18-linker-Sequence optimized ^{Pm/Ra} <i>rdnI</i> (<i>Proteus rdnI</i> with <i>Rothia</i> conserved motif insert) with a C-terminal StreptII with the <i>lac</i> promoter (Col E1 origin, Amp resistance)	gAK026 was recombined into pAK075 using restriction digest
pAK094	T18-linker-Sequence optimized ^{Pm/Pj} <i>rdnI</i> (<i>Proteus rdnI</i> with <i>Prevotella</i> conserved motif insert) with the <i>lac</i> promoter (Col E1 origin, Amp resistance)	gAK028 was recombined into pAK075 using restriction digest

Table 3 (continued)

pAK095	T18-linker-Sequence optimized ^{Pm/Pf} <i>rdnI</i> (<i>Proteus rdnI</i> with <i>Pseudomonas</i> conserved motif insert) with the <i>lac</i> promoter (Col E1 origin, Amp resistance)	gAK027 was recombined into pAK075 using restriction digest
pAK096	T18-linker- ^{Pj/Pm} <i>rdnI</i> (<i>Prevotella rdnI</i> with <i>Proteus</i> conserved motif insert) with the <i>lac</i> promoter (Col E1 origin, Amp resistance)	gAK029 was recombined into pAK075 using restriction digest
pAK097	T18-linker- ^{Pf/Pm} <i>rdnI</i> (<i>Pseudomonas rdnI</i> with <i>Proteus</i> conserved motif insert) with a C-terminal StrepII with the <i>lac</i> promoter (Col E1 origin, Amp resistance)	gAK030 was recombined into pAK075 using restriction digest
pAK064	Sequence optimized <i>rdnI</i> _{mut} -StrepII with the <i>fla</i> promoter, (pBBR1 origin, Kan resistance)	gAK025 was recombined into pAK043 using restriction digest
pAK085	T18-linker-Sequence optimized <i>rdnI</i> _{mut} -StrepII with the <i>lac</i> promoter (Col E1 origin, Amp resistance)	gAK025 was recombined into pAK075 using restriction digest

Table 3 (continued)

In vitro DNase assay

RdnE proteins were produced using the New England Biolabs PURExpress *In Vitro* Protein Synthesis Kit (New England BioLabs Inc., Ipswich MA). Template DNA contained the *rdnE* gene and required elements specified by the PURExpress kit. We adapted this protocol from prior *in vitro* DNA-degradation assays⁶¹. Reactions were performed with 250 ng of template DNA and incubated at 37°C for two hours (no template DNA added to negative control reaction). The protein amount was determined using an anti-FLAG western blot with a known gradient of FLAG-BAP (2.5, 5, 10, and 20 ng). Synthesized protein (2.5, 5, and 10 ng) was added to 0.5 µg of lambda DNA (methylated and unmethylated), 5 µL of New England Biolabs Buffer 3.1, and up to a final volume of 25 µL. For plasmid DNase assays, 10 ng of synthesized protein was added to 250 ng of circular or linear plasmid DNA (pids_{BB})⁶². This reaction was incubated for one hour at 37°C, then Proteinase K (New England Biolabs Inc., Ipswich MA) was added and incubated for an additional 15 minutes at 37°C. The reaction was then run on a 1% agarose gel for analysis.

E. coli liquid growth and viability assays

Overnight cultures were grown at 37°C in a shaking incubator in LB broth with appropriate antibiotics. Cultures were normalized to an optical density at 595 nm (OD₅₉₅) of 1 and diluted 1:100 into LB broth with 35 µg/mL kanamycin, with and without 200 nM anhydrotetracycline (aTc). Samples were analyzed for OD₅₉₅ every thirty minutes for 16 hours in a 96-well plate using a TECAN. Other samples were incubated at 37°C for 6 hours while rocking. At indicated time points, 100 µL of sample was removed, diluted, and then plated on fresh LB agar plates to measure colony forming units per mL (CFU) after overnight growth at 37°C using standard protocols.

Microscopy

We performed microscopy on *P. mirabilis* strain *idrD*::Tn5 (Cm^R) (also called, *idrD*^{*}), which has a transposon insertion to disrupt *rdnE* and *rdnI* expression¹⁹, carrying either vector pBBR1-NheI or pDS0002 (producing RdnE) and on *E. coli* carrying either pBBR1-NheI, pDS0002, or pDS0048 (producing RdnE_{D39A}). *P. mirabilis* cells were normalized to OD₅₉₅ of 0.1 after overnight growth in LB broth supplemented with kanamycin. Cells were inoculated onto CM55 swarm pads containing 10 µg/mL DAPI and 10 nM aTc and grown in humidified chambers at 37°C. Images were taken at five and six hours after growth. From overnight cultures, *E. coli* cells were grown in LB broth plus kanamycin until mid-logarithmic phase and then mounted directly onto glass slides. Glass coverslips were sealed with nail polish. For all microscopy, we captured phase contrast and DAPI (150 ms exposure) images using a Leica DM5500B microscope (Leica Microsystems, Buffalo Grove IL) and CoolSnap HQ CCD camera (Photometrics, Tucson AZ) cooled to -20°C. MetaMorph version 7.8.0.0 (Molecular Devices, Sunnyvale CA) was used for image acquisition.

Sequence Optimized RdnI

The *P. mirabilis rdnI-StrepII* sequence was difficult to genetically engineer due to its low GC% content (23%). As such, we engineered the sequence to have a higher GC%, called “Sequence optimized (SO) *Proteus* RdnI-StrepII” without changes to its amino acid sequence. The change to the nucleotide sequence did not affect the construct’s ability to offer protection to a vulnerable strain (Figure 4B).

Swarm competition assay

The swarm competition (territoriality) assay was adapted from Wenren et. al. 2013¹⁹. 5-mL cultures were grown in LB broth with appropriate antibiotics overnight in a 37°C rocking incubator. Overnight cultures were normalized to an OD₅₉₅ of 1. For the competition samples, the strains were mixed 1:1. 2 µL of each sample were inoculated onto CM55 agar with the appropriate antibiotic. Plates were incubated at 37°C for 22 hours and then photographed and assessed for

boundary formation. All RdnI-producing strains contained a low-copy vector with the *rdnI* gene under the *fla* constitutive promoter; see **Figure 4 - Figure supplement 2** [for](#) relative protein production.

BACTH assay

The vectors are described in Battesti and Bouveret 2012³⁵ [with](#) an added linker region between the T25 or T18 fragments and multiple cloning sites. BTH101 cultures were grown at 30°C overnight in LB broth with kanamycin and carbenicillin. 10 µL of the overnight culture were inoculated onto LB agar with kanamycin, carbenicillin, 1 mM IPTG, and 0 or 40 µg/mL of X-gal (Thermo Fisher, Waltham MA), and grown at 30°C for 24 hours. Color was amplified by an additional 24 hours at 4°C, and then samples were imaged.

FLAG co-immunoprecipitation assays

The protocol was adapted from Cardarelli et. al. 2015² [E. coli](#) cells were harvested from LB broth, grown for either 3 hours after induction with 200 nM aTc at 37°C or 16 to 20 hours at 16°C after induction with 1 mM IPTG. Cells were spun down into pellets using centrifugation and then flash frozen in liquid nitrogen. RdnE-containing samples were lysed in 50mM Tris pH 7.4, 150mM NaCl, and 1x Protease Inhibitor Cocktail (Selleck Chemicals LLC, Houston TX), via bead bashing for 20 minutes at 4°C. RdnI-containing samples were lysed in 100 mM Tris-HCl pH 8, 180 mM NaCl, and 1x Protease Inhibitor Cocktail via 10× 10 second sonication pulses. The soluble fraction for both samples was obtained after centrifugation at 15,000 rpm for 15 minutes. FLAG epitope-containing samples were incubated with prepared resin for two hours at 4°C. The resin was then washed twice (50 mM Tris pH 7.4, 150 mM NaCl, and 1% Tween-20), incubated with approximately 1 mL of the soluble fraction of the RdnI-StrepII-containing samples for another 2 hours, and washed thrice more. The protein was finally eluted with 50 µL of 300 ng/µL 3x FLAG peptide (Sigma-Aldrich, St. Louis, MO) for 45 minutes at 4°C. Sample buffer (63 mM Tris pH 6.8, 2% Sodium Dodecyl Sulfate, 10% glycerol, 5% 2-Mercaptoethanol) was added to samples, boiled at 95°C for 10 minutes, and frozen at -80°C.

P. mirabilis swarm cell protein expression

The protocol was adapted from Cardarelli et. al. 2015² [P. mirabilis](#) cells were harvested from CM55 swarm plates, grown overnight at 37°C. Cells were washed twice in LB broth, spun down by centrifugation, and flash frozen in liquid nitrogen. Cells were lysed in 100 mM Tris-HCl pH 8, 180 mM NaCl, and 1x Protease Inhibitor Cocktail via 10x via bead bashing for 20 minutes at 4°C. The whole cell extract was obtained after centrifugation at 6,000 g for 15 minutes. The soluble fraction for both samples was obtained after subsequent centrifugation at 15,000 rpm for 15 minutes. Sample buffer (63 mM Tris pH 6.8, 2% Sodium Dodecyl Sulfate, 10% glycerol, 5% 2-Mercaptoethanol) was added to samples, boiled at 95°C for 10 minutes, and then immediately used for SDS-Page and Western blotting.

SDS-Page and Western blotting

The protocol was adapted from Cardarelli et. al. 2015² [Protein](#) samples were separated by gel electrophoresis using 13% Tris-Tricine polyacrylamide gels and either transferred to a 0.45-µm nitrocellulose membrane (Bio-Rad Laboratories, Hercules CA) or stained with Coomassie blue (Bio-Rad Laboratories, Hercules CA). Western blot membranes were probed with primary antibody (either 1:4000 rabbit anti-FLAG [Sigma Aldrich, St. Louis MO] or 1:4000 or 1:2000 mouse anti-StrepII [Genscript, Piscataway NJ]) for 1 hour at room temperature or overnight at 4°C and with secondary antibody (either 1:5000 goat anti-rabbit or anti-mouse respectively conjugated to horseradish peroxidase (HRP) [KPL, Inc., Gaithersburg MD]) for 30 minutes for co-IPs or 1 hour for *P. mirabilis* protein expression at room temperature. Samples were finally visualized using Immun-Star HRP substrate kit (Bio-Rad Laboratories, Hercules CA) and the ChemiDoc XRS system (Bio-Rad Laboratories, Hercules, CA). TIFF files were analyzed on Fiji (ImageJ, Madison, WI).

Bioinformatics search for RdnE and RdnI homologs

A BLAST³⁷ search of the *P. mirabilis* RdnE protein sequence revealed seven RdnE homologs from a variety of species. The downstream genes of these RdnE homologs were identified using the DOE Joint Genome Institute (JGI) Integrated Microbial Genomes and Microbiomes (IMG/M). The seven RdnE and RdnI amino acid sequences were separately aligned with MUSCLE using Jalview⁶³. These alignments were then used as seeds for a second homology search using HMMER search (HmmerWeb version 2.41.2) and the Ensembl Database⁶⁴. The two data sets were then compared for genomes that contained both *rdnE* and *rdnI* genes next to one another within their respective genomes. Any EI pairs that contained disrupted PD-(D/E)XK motifs within their RdnE sequence were removed.

Gene neighborhood and primary conservation analyses

Gene neighborhoods were obtained using JGI's IMG/M Neighborhood viewer and then redrawn using Adobe Illustrator (Adobe Inc., 2022). Locations of predicted functions are approximate primarily based on the Pfam domain calling by IMG/M.

The final 21 RdnE and RdnI sequences were aligned with MUSCLE using Jalview⁶³. The conserved residues were identified using Jalview and the cartoons were created using Adobe Illustrator (Adobe Inc., 2022). The sequence logo for the RdnI conserved motif was generated with WebLogo⁶⁵ and constrained to only visualize the conserved motif.

Secondary and tertiary structure predictions

Secondary structure predictions of the MUSCLE aligned sequences were determined with Ali2D from the MPI Bioinformatics toolkit^{65,66}. The resulting predictions were made into cartoons manually using Adobe Illustrator (Adobe Inc., 2022). Tertiary structure predictions were done with AlphaFold2³⁹ using Mmseqs2 on Google Colab⁴⁰. Query protein sequences were inputted into the program and then run producing 5 models ranked 1-5. Rank 1 models are shown. pIDDT scores indicate confidence levels for each amino acid position. Structures were analyzed in PyMOL (The PyMOL Molecular Graphics System, Version 2.2.3 Schrödinger, LLC.). The pIDDT graphs are in the supplemental data.

Metagenomic analyses

A sourmash-based approach was used to screen approximately 500,000 public metagenomes stored on NCBI's SRA (<https://github.com/sourmash-bio/2022-search-sra-with-mastiff>) for the presence of the ten genomes shown in **Figure 3A**. Hits with a containment score greater than 0.2 were downloaded for further analysis, representing 9,137 metagenomes. Each metagenome was then mapped with bbmap⁶⁷ against a reference database the 10 *rdnE* and *rdnI* gene sequence pairs with a stringency of 90% (minid = 0.9), along with quality filtering (trim1 = 20, minaveragequality = 10). 70% stringency was also included and resulted in similar results (**Figure 5 - Figure supplement 1**). After mapping, metagenomes were retained if they had (1) a mean coverage greater than 2X, (2) at least one base covered greater than 5X, and (3) more than half of the bases on reference *rdnE-rdnI* sequence receiving coverage. Coverage of other domains, if any, upstream of the C-terminal domain were not considered for subsequent analysis. 2,857 metagenomes met these criteria, of which 2,296 contained *P. mirabilis*, *R. dentocariosa*, *P. jejuni*, or *P. ogarae* sequences and could be confidently assigned to samples obtained from the human gut or oral microbiome or from terrestrial sources. Gene-level coverage in a sample was then summarized as each gene's average nucleotide's coverage. The ratio of *rdnI* to *rdnE* coverage was then calculated for each sample and log10-transformed, and the distribution of ratios was summarized with Python's seaborn kdeplot using a bandwidth of 0.4⁶⁸.

Open Science statement

The sequence files and associated data are archived on an OSF website (<https://osf.io/scb7z/>) and were made publicly available upon publication.

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Reviewer #3 (Public Review):

Summary:

The authors discovered that the RdnE effector possesses DNase activity, and in competition, *P. mirabilis* having RdnE outcompetes the null strain. Additionally, they presented evidence that the RdnI immunity protein binds to RdnE, suppressing its toxicity. Interestingly, the authors demonstrated that the RdnI homolog from a different phylum (i.e., Actinomycetota) provides cross-species protection against RdnE injected from *P. mirabilis*, despite the limited identity between the immunity sequences. Finally, using metagenomic data from human-associated microbiomes, the authors provided bioinformatic evidence that the *rdnE/rdnI* gene pair is widespread and present in individual microbiomes. Overall, the discovery of broad protection by non-cognate immunity is intriguing, although not necessarily surprising in retrospect, considering the prolonged period during which Earth was a microbial battlefield/paradise.

Strengths:

The authors presented a strong rationale in the manuscript and characterized the molecular mechanism of the RdnE effector both in vitro and in the heterologous expression model. The utilization of the bacterial two-hybrid system, along with the competition assays, to study the protective action of RdnI immunity is informative. Furthermore, the authors conducted bioinformatic analyses throughout the manuscript, examining the primary sequence,

predicted structural, and metagenomic levels, which significantly underscore the significance and importance of the EI pair.

Weaknesses:

- (1) The interaction between RdnI and RdnE appears to be complex and requires further investigation. The manuscript's data does not conclusively explain how RdnI provides a "promiscuous" immunity function, particularly regarding the RdnI mutant/chimera derivatives. The lack of protection observed in these cases might be attributed to other factors, such as a decrease in protein expression levels or misfolding of the proteins. Additionally, the transient nature of the binding interaction could be insufficient to offer effective defenses.
- (2) The results from the mixed population competition would benefit from quantitative analysis. The swarm competition assays only yield binary outcomes (Yes or No), limiting the ability to obtain more detailed insights from the data.
- (3) The discovery of cross-species protection is solely evident in the heterologous expression-competition model. It remains uncertain whether this is an isolated occurrence or a common characteristic of RdnI immunity proteins across various scenarios. Further investigations are necessary to determine the generality of this behavior.

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Reviewer #4 (Public Review):

Summary:

Knecht et al. elucidate a Type VI Secretion System (T6SS) effector-immunity pair in *Proteus mirabilis*. They demonstrate that the effector protein RdnE exhibits DNase activity in vitro and induces toxicity when ectopically expressed in cells, the latter being neutralized by the cognate immunity protein RdnI. The authors identify major regions within RdnI necessary for the interaction and neutralization of RdnE. Notably, they report cross-talk where both cognate and non-cognate RdnI proteins can neutralize RdnE, mitigating its fitness advantage in bacterial co-swarm assays. A comprehensive metagenomic analysis revealed an abundance of *rdnI* over *rdnE* genes in most gut samples, suggesting a potential role of *rdnI* in providing a fitness advantage against bacteria encoding for RdnE effector.

Strengths:

The authors successfully combined biochemical and microbiological experiments with bioinformatics analysis to advance the understanding of the T6SS-mediated population dynamics in bacteria. The co-swarm functional assay is of particular interest as it demonstrates how bacterial strains carrying only *rdnI* immunity genes could potentially compete in the same niche with other species armed with toxic *rdnE* effector genes. The manuscript is well-written, and the figures are self-explanatory.

Weaknesses:

- (1) How would the authors explain the discrepancy observed in Figure 4 G and Figure 4 S3 B where two RdnI proteins from *Prevotella* and *Pseudomonas* genera do not bind to RdnE_*Proteus* in BACTH, whereas they co-elute with a RdnE_*Proteus*-FLAG with efficiency comparable to the cross-neutralizing RdnI_*Rothia*? Similarly, the interaction results obtained in BACTH with RdnI truncate (Figure 4E) or chimeric RdnI (Figure 4I, lane 4) could be a result of an overexpressed T18-fusion variant. Alternative in vitro protein binding assay would be beneficial.

(2) Based on the bioinformatic analysis the *Rothia* and *Prevotella* species harboring *rdnE/I* genes co-occurred in 5% of metagenomes tested, suggesting that these bacteria could come into contact. The manuscript would benefit greatly if authors demonstrated that RdnI proteins from *Rothia* or *Prevotella* could cross-neutralize its own and its 'neighbor' RdnE effectors, for example in an *E. coli* viability assay. The cross-neutralizing co-swarming results (Figure 4F) could also be further validated in viability assay as shown in Figure 2 S1.

(3) Little is known about whether RdnE is delivered via T6SS as a full-length protein or as the shorter C-terminal fragment. There is a possibility that immunity proteins could recognize RdnE regions beyond the C-terminal 138 amino acids that authors used in their in vitro assays.

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Reviewer #5 (Public Review):

This work investigates a T6SS effector-immunity pair from *Proteus mirabilis*. The authors make several interesting claims, particularly regarding the mechanism of effector inhibition by the immunity protein. However, it appears that these claims are not fully supported by the evidence provided.

I have read the revised manuscript, the public reviews, and the authors' updated responses to these reviews. In my opinion, the concerns raised by the reviewers remain relevant even after the authors' revisions. Since previous reviews have excellently described the strengths and weaknesses of this work, I will focus on my major concerns:

(1) The authors describe RdnE-RdnI, a T6SS effector-immunity pair from *Proteus mirabilis*. RdnE is actually the C-terminal domain of IdrD, a 1581-amino-acid protein containing PAAR and RHS domains. This work does not provide evidence for T6SS-dependent secretion of the effector, instead supplying references to previous works.

(2) While the authors claim the function of the RdnE domain is unknown, it was previously shown to be evolutionarily related to PoNe and TseV, both of which are known DNA nucleases. Although the authors cite the relevant references, they do not clearly disclose this information.

(3) The authors claim that RdnE contains two different domains: the first is the PD-(D/E)XK domain, and the second, referred to as "region 2," follows it. Unfortunately, no structural evidence is provided to support this claim, not even a predicted model demonstrating that these are indeed separate domains.

(4) One of the major claims made in this work is that RdnI binding to RdnE is not sufficient for RdnE inhibition, suggesting a more sophisticated mechanism. The authors base this theory on differences between the ability of RdnI to bind RdnE (shown using bacterial two-hybrid assays) and the ability to protect against RdnE toxicity in swarm competition assays. Specifically, they show that the first 85 amino acids of RdnI bind to the short RdnE domain in the bacterial two-hybrid assay but do not protect against the full-length effector in the swarm competition assay. They also demonstrate that performing seven mutations in conserved residues in RdnE or replacing parts of RdnI with parts from other RdnI homologs leads to the same phenomenon.

While these findings are interesting and even intriguing, in my opinion, the current evidence does not support their theory. A simple explanation for the differences between the assays is that while the N-terminal domain of RdnI is sufficient for binding to RdnE, inhibition of the active site of RdnE requires binding of a second domain to RdnE. In that sense, it should be

noted that while the authors use co-IP assays to show the interaction between RdnE and full-length RdnI, they do not use it to show the interaction between RdnE and the first 85 amino acids of RdnI.

(5) The authors claim that a "conserved motif" within RdnI plays a role in the inhibition of RdnE. To investigate this, they replace this motif with sequences from several RdnI homologs, demonstrating that in one case, it is possible to exchange these conserved motifs between RdnI homologs that inhibit Proteus RdnE. However, they also show that even if the conserved motif is taken from an RdnI homolog that cannot inhibit Proteus RdnE, the hybrid protein can still protect cells in a swarm competition assay. This result raises concerns regarding the relevance of this conserved motif.

(6) Lastly, regarding the theory that immunity proteins can protect against non-cognate effectors, it appears that the authors based their theory on a single case where RdnI from *Rothia* protected against RdnE from *Proteus*. In my opinion, a more thorough investigation, involving testing many homologs, is needed to substantiate this theory.

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Author response:

The following is the authors' response to the original reviews.

eLife assessment

This work presents valuable information about the specificity and promiscuity of toxic effector and immunity protein pairs. The evidence supporting the claims of the authors is currently incomplete, as there is concern about the methodology used to analyze protein interactions, which did not take potential differences in expression levels, protein folding, and/or transient interaction into account. Other methods to measure the strength of interactions and structural predictions would improve the study. The work will be of interest to microbiologists and biochemists working with toxin-antitoxin and effector-immunity proteins.

We thank the reviewers for considering this manuscript. We agree that this manuscript provides a valuable and cross-discipline introduction to new EI pair protein families where we focus on the EI pair's flexibility and impacts on community structure. As such, we believe we have provided a solid foundation for future studies to examine non-cognate interactions and their possible effects on microbial communities. This, by definition, leaves some areas "incomplete" and, therefore, open for further investigations. While the methods we show do consider potential differences in binding assays, we have more explicitly addressed how "expression, protein folding, and/or transient binding" may play into this expanded EI pair model. We have also tempered the discussion of the proposed model, while also clearly highlighting other published evidence of non-cognate binding interactions between effector and immunity proteins. We have responded to the reviewers' public comments (italicized below).

In this revised manuscript, we have updated the main text, particularly the Discussion section, to include more careful language, explain past research better, and add new references to works showing non-cognate immunity proteins protecting against effectors in other systems. We have also updated the supplemental files with more analyses; the relevant procedures are in the Materials and Methods.

Public Reviews:

Note: Reviewer 1, who appeared to focus on a subset of the manuscript rather than the whole, based their comments on several inaccuracies, which we discuss below. We found the tone in this reviewer's comments to be, at times, inappropriate, e.g., using "harsh" and "simply too drastic" to imply that common structure-function analyses were outside of the field-standard methods. We also note that the reviewer took a somewhat atypical step in reviewing this manuscript by running and analyzing the potential protein-complex data in AlphaFold2 but did not discuss areas of low confidence within that model that may contradict their conclusions. We are concerned their approach muddled valid scientific criticisms with problematic conclusions.

Reviewer #1 (Public Review):

In this manuscript, Knecht, Sirias et al describe toxin-immunity pair from Proteus mirabilis. Their observations suggest that the immunity protein could protect against non-cognate effectors from the same family. They analyze these proteins by dissecting them into domains and constructing chimeras which leads them to the conclusion that the immunity can be promiscuous and that the binding of immunity is insufficient for protective activity.

Strengths:

The manuscript is well written and the data are very well presented and could be potentially interesting. The phylogenetic analysis is well done, and provides some general insights.

Weaknesses:

(1) Conclusions are mostly supported by harsh deletions and double hybrid assays. The later assays might show binding, but this method is not resolutive enough to report the binding strength. Proteins could still bind, but the binding might be weaker, transient, and out-competed by the target binding.

The phrasing of structure-function analyses as "harsh" is a bit unusual, as other research groups regularly use deletions and hybrid studies. Given the known caveats to deletion and domain substitutions, we included point-mutation analyses for both the effector and immunity proteins, as found on lines 105 - 113 and 255 - 261 in the current manuscript. These caveats are also why we coupled the in vitro binding analyses with in vivo protection experiments in two distinct experimental systems (*E. coli* and *P. mirabilis*). Based on this manuscript's introductory analysis (where we define and characterize the genes, proteins, interactions, phylogenetics, and incidences in human microbiomes), the next apparent questions are beyond the scope of this study. Future approaches would include analyzing purified proteins from the effector (E) and immunity (I) protein families using biochemical assays, such as X-ray crystallography, circular dichroism spectroscopy, among others.

Interestingly, most papers in the EI field do not measure EI protein affinity (Jana et al., 2019, Yadav et al., 2021). Notable exceptions are earlier colicin research (Wallis et al., 1995) and a new T6SS EI paper (Bosch et al., 2023) published as we first submitted this manuscript.

(2) While the authors have modeled the structure of toxin and immunity, the toxin-immunity complex model is missing. Such a model allows alternative, more realistic interpretation of the presented data. Firstly, the immunity protein is predicted to bind contributing to the surface all over the sequence, except the last two alpha helices (very high confidence model, iPTM>0.8). The N terminus described by the authors contributes one of the toxin-binding surfaces, but this is not the sole binding site. Most importantly,

other parts of the immunity protein are predicted to interact closer to the active site (D-E-K residues). Thus, based on the AlphaFold model, the predicted mechanism of immunization remains physically blocking the active site. However, removing the N terminal part, which contributes large interaction surface will directly impact the binding strength. Hence, the toxin-immunity co-folding model suggests that proper binding of immunity, contributed by different parts of the protein, is required to stabilize the toxin-immunity complex and to achieve complete neutralization. Alternative mechanisms of neutralization might not be necessary in this case and are difficult to imagine for a DNase.

In response to the reviewer's comment, we again reviewed the RdnE-RdnI AlphaFold2 complex predictions with the most updated version of ColabFold (1.5.2-patch with PDB100 and MMseq2) and have included them at the end of these responses [1].

However, the literature reports that computational predictions of E-I complexes often do not match experimental structural results (Hespanhol et al., 2022, Bosch et al., 2023). As such, we chose not to include the predicted cognate and non-cognate RdnE-I complexes from ColabFold (which uses AlphaFold2) and have not included this data in the revised manuscript. (It is notable that reviewer 1 found the proposed expanded model and research so interesting as to directly input and examine the AI-predicted RdnE-RdnI protein interactions in AlphaFold2.)

Discussion of the prevailing toxin-immunity complex model is in the introduction (lines 45-48) and Figure 5E. Further, there are various known mechanisms for neutralizing nucleases and other T6SS effectors, which we briefly state in the discussion (lines 359 - 361). More in-depth, these molecular mechanisms include active-site blocking (Benz et al., 2012), allosteric-site binding (Kleanthous et al., 1999 and Lu et al., 2014), enzymatic neutralization of the target (Ting et al., 2021), and structural disruption of both the active and binding sites (Bosch et al., 2023). Given this diversity of mechanisms, we did not presume to speculate on the as-of-yet unknown mechanism of RdnI protection. We have expanded discussion of these items in the revised manuscript.

(3) Dissection of a toxin into two domains is also not justified from a structural point of view, it is probably based on initial sequence analyses. The N terminus (actually previously reported as Pone domain in ref 21) is actually not a separate domain, but an integral part of the protein that is encased from both sides by the C terminal part. These parts might indeed evolve faster since they are located further from the active site and the central core of the protein. I am happy to see that the chimeric toxins are active, but regarding the conservation and neutralization, I am not surprised, that the central core of the protein fold is highly conserved. However, "deletion 2" is quite irrelevant - it deletes the central core of the protein, which is simply too drastic to draw any conclusions from such a construct - it will not fold into anything similar to an original protein, if it will fold properly at all.

The reviewer's comment highlights why we turned to the chimera proteins to dissect the regions of RdnE (formerly IdrD-CT), as the deletions could result in misfolded proteins. (We initially examined RdnE in the years before the launch of AlphaFold2.) However, the reviewer is incorrect regarding the N-terminus of RdnE. The PoNe domain, while also a subfamily of the PD-(D/E)XK superfamily, forms a distinct clade of effectors from the PD-(D/E)XK domain in RdnE (formally IdrD-CT) as seen in Hespanhol et al., 2022; this is true for other DNase effectors as well. Many studies analyzing effectors within the PD-(D/E)XK superfamily only focus on the PD-(D/E)XK domain, removing just this domain from the context of the whole protein (Hespanhol et al., 2022; Jana et al., 2019). Of note, in RdnE, this region alone (containing the DNA-binding domain) is insufficient for DNase activity (unlike in

PoNe). We have clarified this distinction in the results section of the current manuscript, visible in figure 2 .

(4) Regarding the "promiscuity" there is always a limit to how similar proteins are, hence when cross-neutralization is claimed authors should always provide sequence similarities. This similarity could also be further compared in terms of the predicted interaction surface between toxin and immunity.

Reviewer 1 points out a fundamental property of protein-protein interactions that has been isolated away from the impacts of such interactions on bacterial community structure. We have provided the whole protein alignments in figure 3 supplemental figure 3, the summary images in Figure 3D, and the protein phylogenetic trees in Figure 3C. We encourage others to consider the protein alignments as percent amino acid sequence similarity is not necessarily a good gauge for protein function and interactions. These data are publicly available on the OSF website associated with this manuscript <https://osf.io/scb7z/>, and we hope the community explores the data there.

In consideration of the enthusiasm to deeply dive into the primary research data, we have included the pairwise sequence identities across the entire proteins here: Proteus RdnI vs. Rothia RdnI: 23.6%; Proteus RdnI vs. Prevotella RdnI: 16.3%, Proteus RdnI vs. Pseudomonas RdnI: 14.6%; Rothia RdnI vs. Prevotella RdnI: 22.4%, Rothia RdnI vs. Pseudomonas RdnI: 17.6%; Prevotella RdnI vs. Pseudomonas RdnI: 19.5%. (As stated in response to reviewer 1 comment 2, we did not find it appropriate to make inferences based on AlphaFold2-predicted protein complexes.)

Overall, it looks more like a regular toxin-immunity couple, where some cross-reactions with homologues are possible, depending on how far the sequences have deviated. Nevertheless, taking all of the above into account, these results do not challenge toxin-immunity specificity dogma.

In this manuscript, we did not intend to dismiss the E-I specificity model but rather point out its limitations and propose an important expansion of that model that accounts for cross-protection and survival against attacks from other genera. We agree that it is commonly considered that deviations in amino acid sequence over time could result in cross-binding and protection (see lines 364-368). However, the impacts of such cross-binding on community structure, bacterial survival, and strain evolution were rarely addressed in prior literature, with exceptions such as in Zhang et al., 2013 and Bosch et al., 2023 among others. One key insight we propose and show in this manuscript is that cross-binding can be a fitness benefit in mixed communities; therefore, it could be selected for evolutionarily (lines 378-380), even potentially in host microbiomes.

Reviewer #2 (Public Review):

Summary:

The manuscript by Knecht et al entitled "Non-cognate immunity proteins provide broader defenses against interbacterial effectors in microbial communities" aims at characterizing a new type VI secretion system (T6SS) effector immunity pair using genetic and biochemical studies primarily focused on Proteus mirabilis and metagenomic analysis of human-derived data focused on Rothia and Prevotella sequences. The authors provide evidence that RdnE and RdnI of Proteus constitute an E-I pair and that the effector likely degrades nucleic acids. Further, they provide evidence that expression of non-cognate immunity derived from diverse species can provide protection against RdnE intoxication. Overall, this general line of investigation is underdeveloped in the T6SS field and conceptually appropriate for a broad audience journal. The paper is well-written

and, aside from a few cases, well-cited. As detailed below however, there are several aspects of this paper where the evidence provided is somewhat insufficient to support the claims. Further, there are now at least two examples in the literature of non-cognate immunity providing protection against intoxication, one of which is not cited here (Bosch et al PMID 37345922 - the other being Ting et al 2018). In general therefore I think that the motivating concept here in this paper of overturning the predominant model of interbacterial effector-immunity cognate interactions is oversold and should be dialed back.

We agree that analyses focusing on flexible non-cognate interactions and protection are underdeveloped within the T6SS field and are not fully explored within a community structure. These ideas are rapidly growing in the field, as evidenced by the references provided by the reviewer. As stated earlier, we did not intend to overturn the prevailing model but rather have proposed an expanded model that accounts for protection against attacks from foreign genera.

Strengths:

One of the major strengths of this paper is the combination of diverse techniques including competition assays, biochemistry, and metagenomics surveys. The metagenomic analysis in particular has great potential for understanding T6SS biology in natural communities. Finally, it is clear that much new biology remains to be discovered in the realm of T6SS effectors and immunity.

Weaknesses:

The authors have not formally shown that RdnE is delivered by the T6SS. Is it the case that there are not available genetics tools for gene deletion for the BB2000 strain? If there are genetic tools available, standard assays to demonstrate T6SS-dependency would be to interrogate function via inactivation of the T6SS (e.g. by deleting tssC).

Our research group showed that the T6SS secretes RdnE (previously IdrD) in Wenren et al., 2013 (cited in lines 71-73). We later confirmed T6SS-dependent secretion by LC-MS/MS (Saak et al., 2017).

For swarm cross-phyla competition assays (Figure 4), at what level compared to cognate immunity are the non-cognate immunity proteins being expressed? This is unclear from the methods and Figure 4 legend and should be elaborated upon. Presumably these non-cognate immunity proteins are being overexpressed. Expression level and effector-to-immunity protein stoichiometry likely matters for interpretation of function, both in vitro as well as in relevant settings in nature. It is important to assess if native expression levels of non-cognate cross-phyla immunity (e.g. Rothia and Prevotella) protect similarly as the endogenously produced cognate immunity. This experiment could be performed in several ways, for example by deleting the RdnE-I pair and complementing back the Rothia or Prevotella RdnI at the same chromosomal locus, then performing the swarm assay. Alternatively, if there are inducible expression systems available for Proteus, examination of protection under varying levels of immunity induction could be an alternate way to address this question. Western blot analysis comparing cognate to non-cognate immunity protein levels expressed in Proteus could also be important. If the authors were interested in deriving physical binding constants between E and various cognate and non-cognate I (e.g. through isothermal titration calorimetry) that would be a strong set of data to support the claims made. The co-IP data presented in supplemental Figure 6 are nice but are from E. coli cells overexpressing each protein and do not fully address the question of in vivo (in Proteus) native expression.

P. mirabilis strain ATCC29906 does not encode the *rdnE* and *rdnI* genes on the chromosome (NCBI BioSample: SAMN00001486) (line 151). Production of the RdnI proteins, including the cognate *Proteus* RdnI, comes from equivalent transgenic expression vectors. Specifically, the *rdnI* genes were expressed under the *flaA* promoter in *P. mirabilis* strain ATCC29906 (Table 1) for the swarm competition assays found in Figure 2C and Figure 4. This promoter results in constitutive expression in swarming cells (Belas et al., 1991; Jansen et al., 2003). In the revised manuscript, figure 4 Supplement Figure 2 shows the relative RdnI protein levels in these strains; we also clarified the expression constructs in the text (see reviewer 3, comment 1).

Lines 321-324, the authors infer differences between E and I in terms of read recruitment (greater abundance of I) to indicate the presence of orphan immunity genes in metagenomic samples (Figure 5A-D). It seems equally or perhaps more likely that there is substantial sequence divergence in E compared to the reference sequence. In fact, metagenomes analyzed were required only to have "half of the bases on reference E-I sequence receiving coverage". Variation in coverage again could reflect divergent sequence dipping below 90% identity cutoff. I recommend performing metagenomic assemblies on these samples to assess and curate the E-I sequences present in each sample and then recalculating coverage based on the exact inferred sequences from each sample.

This comment raises the challenges with metagenomic analyses. It was difficult to balance specificity to a particular species' DNA sequence with the prevalence of any homologous sequence in the sample. Given the distinction in binding interactions among the examined four species, we opted to prioritize specificity, accepting that we were losing access to some *rdnE* and *rdnI* sequences in that decision. We chose a 90% identity cutoff, which, through several in silico controls, ensured that each sequence we identified was the *rdnE* or *rdnI* gene from that specific species. For the Version of Record, we have included analysis with a 70% cutoff in the supplemental information to try to account for sequence divergence by lowering the identity cutoffs as suggested. The data from the 70% identity cutoff was consistent with the original data from the 90% identity cutoff.

A description of gene-level read recruitment in the methods section relating to metagenomic analysis is lacking and should be provided.

Noted. We included the raw code and sequences on the OSF website associated with this manuscript <https://osf.io/scb7z/>.

Reviewer #3 (Public Review):

Summary:

*The authors discovered that the RdnE effector possesses DNase activity, and in competition, P. mirabilis having RdnE outcompetes the null strain. Additionally, they presented evidence that the RdnI immunity protein binds to RdnE, suppressing its toxicity. Interestingly, the authors demonstrated that the RdnI homolog from a different phylum (i.e., Actinomycetota) provides cross-species protection against RdnE injected from P. mirabilis, despite the limited identity between the immunity sequences. Finally, using metagenomic data from human-associated microbiomes, the authors provided bioinformatic evidence that the *rdnE/rdnI* gene pair is widespread and present in individual microbiomes. Overall, the discovery of broad protection by non-cognate immunity is intriguing, although not necessarily surprising in retrospect, considering the prolonged period during which Earth was a microbial battlefield/paradise.*

Strengths:

The authors presented a strong rationale in the manuscript and characterized the

molecular mechanism of the RdnE effector both in vitro and in the heterologous expression model. The utilization of the bacterial two-hybrid system, along with the competition assays, to study the protective action of RdnI immunity is informative. Furthermore, the authors conducted bioinformatic analyses throughout the manuscript, examining the primary sequence, predicted structural, and metagenomic levels, which significantly underscore the significance and importance of the EI pair.

Weaknesses:

(1) The interaction between RdnI and RdnE appears to be complex and requires further investigation. The manuscript's data does not conclusively explain how RdnI provides a "promiscuous" immunity function, particularly concerning the RdnI mutant/chimera derivatives. The lack of protection observed in these cases might be attributed to other factors, such as a decrease in protein expression levels or misfolding of the proteins. Additionally, the transient nature of the binding interaction could be insufficient to offer effective defenses.

Yes, we agree with the reviewer and hope that grant reviewers' share this colleague's enthusiasm for understanding the detailed molecular mechanisms of RdnE-RdnI binding across genera. In the revised manuscript, we have continued to emphasize such caveats as the next frontier is clearly understanding the molecular mechanisms for RdnI cognate or non-cognate protection. In the revised manuscript, figure 4 Supplement Figure 2 shows the RdnI protein levels; we also clarified the expression constructs in the text (see reviewer 2, comment 2).

(2) The results from the mixed population competition lack quantitative analysis. The swarm competition assays only yield binary outcomes (Yes or No), limiting the ability to obtain more detailed insights from the data.

The mixed swam assay is needed when studying T6SS effectors that are primarily secreted during *Proteus*' swarming activity (Saak et al. 2017, Zepeda-Rivera et al. 2018). This limitation is one reason we utilize in vitro, in vivo, and bioinformatic analyses. Though the swarm competition assay yields a binary outcome, we are confident that the observed RdnI protection is due to interaction with a trans-cell RdnE via an active T6SS. By contrast, many manuscripts report co-expression of the EI pair (Yadev et al., 2021, Hespanhol et al., 2022) rather than secreted effectors, as we have achieved in this manuscript.

(3) The discovery of cross-species protection is solely evident in the heterologous expression-competition model. It remains uncertain whether this is an isolated occurrence or a common characteristic of RdnI immunity proteins across various scenarios. Further investigations are necessary to determine the generality of this behavior.

We agree, which is why we submitted this paper as a launching point for further investigations into the generality of non-cognate interactions and their potential impact on community structure.

Comments from Reviewing Editor:

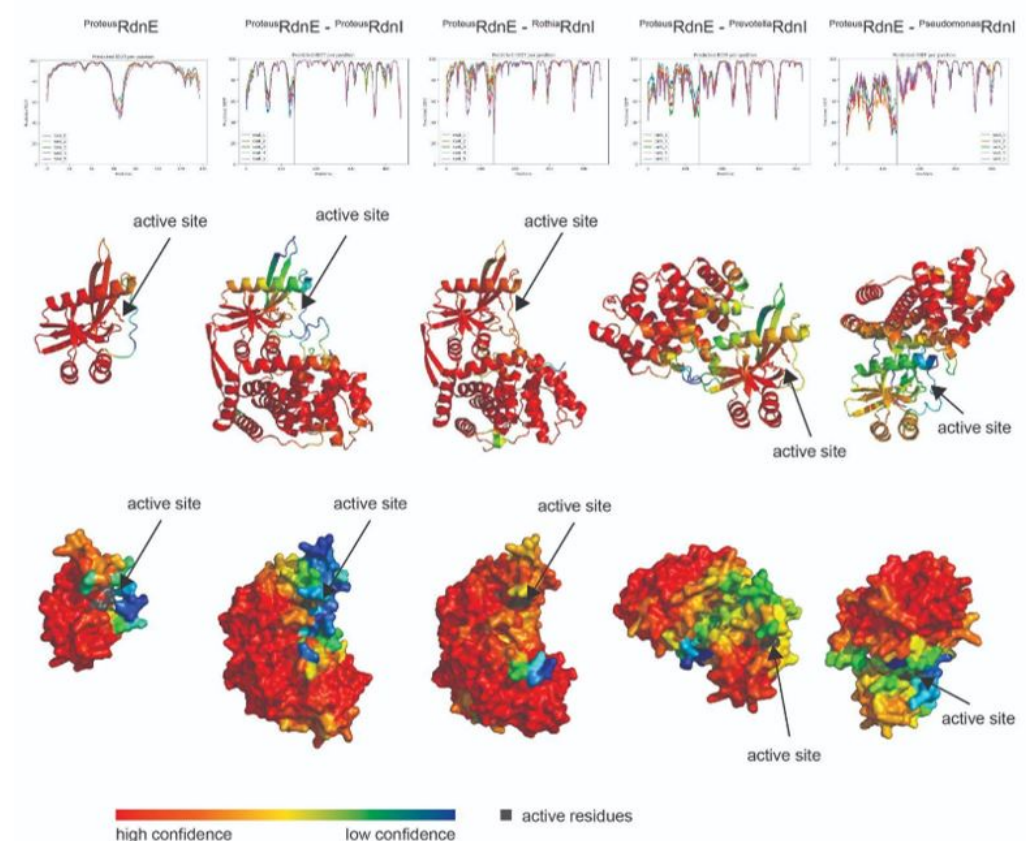
- *In addition to the references provided by reviewer#2, the first manuscript to show non-cognate binding of immunity proteins was Russell et al 2012 (PMID: 22607806).*
- *IdrD was shown to form a subfamily of effectors in this manuscript by Hespanhol et al 2022 PMID: 36226828 that analyzed several T6SS effectors belonging to PDDExK, and it should be cited.*

We appreciate that the reviewer and eLife staff pointed out missed citations. We have incorporated these studies and cited them in the revised manuscript.

[1]

The *Proteus* RdnE in complex with either the *Prevotella* or *Pseudomonas* RdnI showed low confidence at the interface (pIDDT ~50-70%); this AI-predicted complex might support the lack of binding seen in the bacterial two-hybrid assay. On the other hand, the *Proteus* and *Rothia* RdnI N-terminal regions show higher confidence at the interface with RdnE. Despite this, the C-terminus of the *Proteus* RdnI shows especially low confidence (pIDDT ~50%) where it might interact near RdnE's active site (as suggested by reviewer 1). Given this low confidence and the already stated inaccuracies of AI-generated complexes, we would rather wait for crystallization data to inform potential protection mechanisms of RdnI.

Author response image 1.



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